EPILEPSY DIGITAL TWIN REPORT · INTEGRATED NEUROLOGICAL DECISION SUPPORT

Neuro Digital Twin Integrated Clinical Report

An integrated clinical assessment combining follow-up visits, seizure history, EEG, ECG, CT/MRI imaging, laboratory results, and pharmacogenomic testing for Focal impaired-awareness seizures.

Patient & Clinical Context

11/11 data categories included · generated 2026-10-02T17:19:25

PATIENT	AGE / SEX	PRIMARY DIAGNOSIS	DIAGNOSED SINCE
John Doe Patient ID SYN-100482	**34 years · Male**	**Focal impaired-awareness seizures** Secondary diagnosis: Secondary generalized tonic-clonic seizures	**12 May 2021**

CURRENT CLINICAL STATUS
Stable outpatient epilepsy state with persistent breakthrough seizure risk

CURRENT THERAPY
Levetiracetam (Keppra) 1000 mg BID · Lamotrigine 150 mg BID

REPORTING PERIOD
Data reviewed: 01 Jan – 30 Jun 2026 (~6 months)
Clinic visit notes cover 01 Jan 2026 to 02 Mar 2026; laboratory, imaging, EEG/ECG, and questionnaire data continue through 30 Jun 2026.

5	12	6	6	12	35	12
Clinical Visits	CBC Reports	CT Studies	MRI Series	ECG Sessions	EEG Recordings	Questionnaires

01 Executive Summary

Across five clinical-note visits from 2026-01-01 to 2026-03-02, the patient has chronic focal epilepsy with persistent breakthrough seizures on levetiracetam and lamotrigine. The clinician repeatedly documented reasonably stable seizure control and favorable quality-of-life trajectory, while seizure burden, sleep quality, trigger exposure, aura symptoms, and medication tolerability continued to fluctuate.

67%	Moderate	Elevated	10 (7 high)
Overall Health Score	Overall Clinical Risk	EEG-Derived Seizure Risk	Genetic Safety Flags

Domain the patient rates worst on questionnaires: Epileptic Triggers. See the Digital Twin Risk Dashboard (Section 13) for the full risk breakdown.

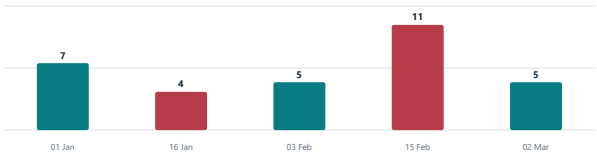
02 6-Month Timeline

- **01 JAN 2026**
  Routine epilepsy follow-up with stable clinician impression — Contributes to longitudinal assessment of seizure burden, triggers, sleep, and tolerability.
- **16 JAN 2026**
  Seizure clustering documented during follow-up — Raises near-term seizure-monitoring priority despite otherwise stable clinician impression.
- **03 FEB 2026**
  CBC/LFT safety review documented — Supports medication-safety monitoring without a documented abnormal lab signal.
- **15 FEB 2026**
  Highest recorded seizure burden in clinical-note window — 11 reported episodes make this the peak burden visit.
- **02 MAR 2026**
  Routine epilepsy follow-up with stable clinician impression — Contributes to longitudinal assessment of seizure burden, triggers, sleep, and tolerability.
- **10 SEP 2026**
  Pharmacogenomic (PGx) panel reported — 76 variants analyzed across 47 medications; see Pharmacogenomics.

03 Seizure Trends

Clinical-note episode counts, clustering, and duration

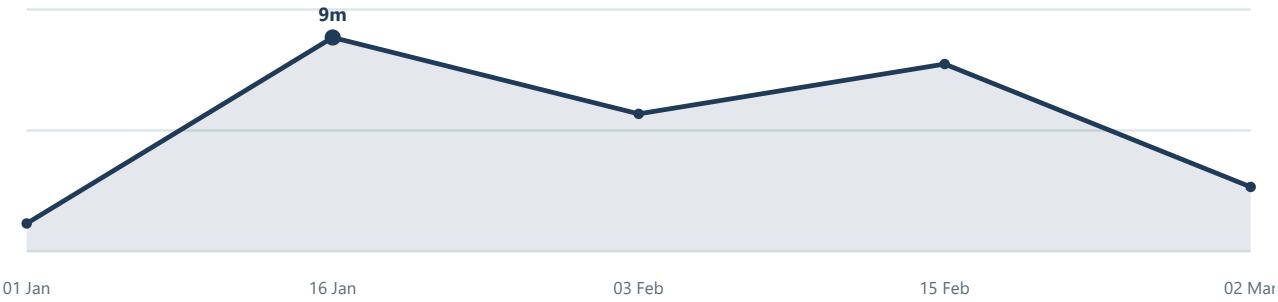
Reported Episodes per Visit



Red bars mark visits with a high-importance clinical event.

OVERALL PATTERN	Breakthrough seizure burden fluctuated across visits without a sustained monotonic worsening pattern.
HIGHEST BURDEN	11 episodes on 15 Feb 2026
LOWEST BURDEN	4 episodes on 16 Jan 2026
CLUSTERING	Yes
CLINICAL INTERPRETATION	Seizure risk remains clinically relevant because events persist on dual therapy and clustering was documented once.

Average Seizure Duration per Visit (minutes)



04 Aura & Triggers

Prodromal patterns and patient-reported trigger exposure

Persistent Aura / Prodromal Patterns

Deja vu frequency · Epigastric rising sensation · Feeling of derealization/detachment · Metallic taste in mouth · Racing heartbeat prior to episode · Sudden dizziness before seizure onset · Sudden intense anxiety or fear · Tingling or numbness sensations · Unusual smells (olfactory aura) · Visual distortions or flashing lights

Aura and prodromal symptoms remain useful warning signals for episode tracking.

Aura & Prodromal Symptoms — Questionnaire Domain

Latest score **71.7/100** (first 66.7, trend increasing)

- Did you experience an epigastric "rising" sensation in the stomach: **Frequent**
- Frequency of sudden, unprovoked psychological anxiety or panic feelings (ictal fear): **Rare**
- Frequency of sudden Deja vu or Jamais vu experiences: **Never**
- Did you notice an anomalous metallic, chemical, or bitter taste in your mouth: **Rare sensory distortion**
- Did you experience unprovoked olfactory distortions (strange, non-existent smells): **Never**

Epileptic Triggers — Lowest-Scoring Domain

Latest score **56.7/100** (min 56.7, trend stable)

- Rate your average psychological stress and workload level this month: **High/Distressing**
- Did you experience a delayed or completely missed dose of your anti-seizure medication: **1 missed dose**
- Exposure to flickering/flashing lights, high-contrast patterns, or prolonged screen glare: **Prolonged exposure**
- Daily average volume of dietary caffeine intake (coffee, tea, energy drinks): **1 cup**
- Any consumption of alcoholic beverages or exposure to hangover states this month: **Occasional drinking**

05 Medication Response

Current regimen, tolerability, and reported side effects

Current Regimen

DRUG	DOSE
Levetiracetam (Keppra)	1000 mg BID
Lamotrigine	150 mg BID

EFFECTIVENESS	Partial response
TOLERABILITY	Generally tolerated with intermittent dizziness and variable side-effect burden
DOCUMENTED TOXICITY	No
CLINICAL INTERPRETATION	Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.

Persistent Side Effects (clinical notes)

Memory fog post-dose | Tremor in hands | Weight change since last visit | Gum hypertrophy | Mood irritability | Nausea after taking medication | Skin rash or itching | Daytime drowsiness | Dizziness/lightheadedness | Unsteady gait (ataxia)

Medication Side Effects — Questionnaire Domain

Latest score **78.3/100** (first 68.3, trend increasing)

- Rate the severity of daytime drowsiness or somnolence: **Mild (Manageable)**
- Experience of dizziness, vertigo, or lightheadedness after taking medication: **Never**
- Presence of physical instability, unsteadiness, or clumsy movements (ataxia): **Slight imbalance**
- Frequency of unprovoked irritability, anger, or mood volatility: **Mild irritability**
- Signs of dermatological distress or unexplained skin rashes: **Clear skin**

06 Drug Interactions

2 current medication(s) reconciled · Flockhart Cytochrome P450 Drug Interaction Table (curated-snapshot-2024)

DDI & Patient-Specific Therapy Assessment — Summary

Overall: 1 clinically relevant findings | 2 unresolved finding(s).

3	0	1	2
Total Findings Assessed	Clinically Significant	Resolved — No Concern	Unresolved

MEDICATION	FLAG	KEY FINDING	PATIENT IMPACT	EVIDENCE SOURCE
Lamotrigine	UNRESOLVED	Therapeutic evidence present, but clinical effectiveness remains incomplete.	Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.	Patient pharmacogenomic findings; Laboratory findings; Clinical record; EEG findings
Levetiracetam (Keppra)	UNRESOLVED	Therapeutic evidence present, but clinical effectiveness remains incomplete.	Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.	Patient pharmacogenomic findings; Laboratory findings; Clinical record; EEG findings

Key clinical interpretation: These flags identify findings that require ongoing clinical monitoring, not an automatic medication change. Each finding should be interpreted alongside the patient's actual treatment response, adverse effects, and the clinical/laboratory findings documented elsewhere in this report -- a resolved or unresolved flag is not itself a prescribing instruction.

Monitoring / Clinician Review Checklist

- Medication reconciliation (dose, frequency, and route consistent across all source records)
- Actual treatment response (seizure/symptom control on the current regimen)
- Relevant adverse effects reported since the last review
- Therapeutic drug levels, when indicated and available
- Relevant laboratory and clinical parameters (e.g. renal, hepatic, hematological)

Medication Reconciliation

DRUG	DOSE	FLAGS
Lamotrigine	150.0 mg BID	—
Levetiracetam (Keppra)	1000.0 mg BID	—

No conflicting doses found for the same drug across the reconciled source data.

Drug Interaction Flag Sheet — Extracted Drug List

Positive Impact = curated/literature evidence explicitly describes a favorable or well-tolerated combination; **Negative Impact** = a real curated pharmacokinetic mechanism, a pharmacodynamic-risk rule, or literature language explicitly describing a risk. A pair with no evidence, or whose evidence never clearly resolved either way, is not listed here at all -- it is never guessed into either column (see Known Limitations).

Screened 26 drug(s) across 325 unique pairs; 80 pair(s) resolved clearly to Positive or Negative and are shown below. 245 pair(s) had no evidence, or evidence that did not clearly resolve either way, and are not listed here -- never shown as a negative/no-concern finding (see Known Limitations).

ARIPIPRAZOLE

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Aripiprazole + Carbamazepine** **Findings:** Aripiprazole's DailyMed label states: "Strong CYP3A4 Inducers (e.g., carbamazepine, rifampin) Concomitant use of aripiprazole and carbamazepine decreased the exposure of aripiprazole compared to the use of aripiprazole alone [see Clinical Pharmacology" **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed **Aripiprazole + Fluoxetine** **Findings:** Fluoxetine, listed as a strong CYP2D6 inhibitor in the aripiprazole label, increases aripiprazole exposure when used concomitantly compared with aripiprazole alone. The label recommends reducing the aripiprazole dosage when it is administered with a strong CYP2D6 inhibitor such as fluoxetine. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed **Aripiprazole + Paroxetine** **Findings:** Paroxetine is listed as a strong CYP2D6 inhibitor whose concomitant use with aripiprazole increased aripiprazole exposure compared with aripiprazole alone.

POSITIVE IMPACT	NEGATIVE IMPACT
	The DailyMed label recommends reducing the aripiprazole dosage when it is administered with a strong CYP2D6 inhibitor such as paroxetine. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed
	Aripiprazole + Venlafaxine **Findings:** The FDA-label-derived evidence states that no dosage adjustment is necessary for CYP2D6 substrates, including venlafaxine, when co-administered with aripiprazole. This is presented as an aripiprazole prescribing-label drug-interaction statement, with no additional strong caution language or specific adverse outcome stated for this pair. **Evidence:** DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** General web search (Tavily)

ATOMOXETINE

POSITIVE IMPACT	NEGATIVE IMPACT
Atomoxetine + Olanzapine **Findings:** Open-label treatment with atomoxetine plus olanzapine for 10 weeks was associated with statistically significant improvement in ADHD symptoms and overt aggression in children and adolescents with ADHD and comorbid disruptive behavior disorder. In 11 patients ages 10–18, symptoms fell by 33% on the ADHD-RS-I and MOAS; 73% responded for ADHD and 55% for aggression, and both medications were generally well tolerated, although olanzapine was associated with significant weight gain averaging 3.9 kg. **Evidence:** External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** PubMed	**Atomoxetine + Paroxetine** **Findings:** Paroxetine may increase atomoxetine exposure because paroxetine is a CYP2D6 inhibitor and atomoxetine is listed as a CYP2D6 substrate in the label's Drug Interactions section. The label recommends decreasing the dosage of the CYP2D6 substrate if needed during concomitant paroxetine use, and notes that an increase in the substrate dose may be needed if paroxetine is discontinued. **Evidence:** DailyMed label (Drug Interactions section); External evidence (PubMed); General web search (Tavily) **Hyperlink:** DailyMed
Atomoxetine + Valproic Acid **Findings:** Methylphenidate or atomoxetine for 2 weeks also improved the prenatal VPA-induced decrease in dendritic spine density in the prefrontal cortex. The effects of these drugs on behavior **Evidence:** External evidence (PubMed) **Hyperlink:** PubMed	**Atomoxetine + Sertraline** **Findings:** Sertraline hydrochloride may increase atomoxetine exposure because sertraline is a CYP2D6 inhibitor and atomoxetine is listed among CYP2D6 substrates in the label. The DailyMed interaction section recommends decreasing the dosage of a CYP2D6 substrate if needed during concomitant sertraline use, and notes that the substrate dose may need to be increased if sertraline is discontinued. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed

ATORVASTATIN

POSITIVE IMPACT	NEGATIVE IMPACT
Atorvastatin + Nortriptyline **Findings:** Atorvastatin added to nortriptyline significantly reduced migraine attack frequency versus placebo plus nortriptyline after 24 weeks (P = 0.004), improved quality of life at follow-up intervals including week 14 (P = 0.001) and week 24, and did not improve headache intensity. This was a prospective, triple-blind randomized controlled trial in 68 adult migraine patients, with constipation reported as the most common adverse effect in the atorvastatin combination arm. **Evidence:** External evidence (PubMed) **Hyperlink:** PubMed	**Atorvastatin + Carbamazepine** **Findings:** Chronic atorvastatin attenuated carbamazepine's anticonvulsant effect in a mouse maximal electroshock seizure model, increasing carbamazepine's ED50 against maximal electroconvulsions. The study evaluated acute and 7-day statin treatment with antiepileptic drugs in mice and also assessed adverse-effect potential and antiepileptic-drug brain concentrations, but the provided evidence does not state a clinical recommendation for this pair. **Evidence:** External evidence (PubMed) **Hyperlink:** PubMed
	Atorvastatin + Lamotrigine **Findings:** When atorvastatin was administered with lamotrigine compared with when atorvastatin was administered alone, atorvastatin AUC((0-τ)) was within bounds indicating no interaction, whereas C(max) was slightly higher(14%); AUC((0-τ)) and C(max) were 3% and 20% higher, respectively, for 2OH-atorvastatin and 25% and 21% higher, respectively, for 4OH-atorvastatin.When atorvastatin was administered with phenytoin compared with when atorvastatin was administered alone, reductions in AUC((0-τ)) and C(max) were observed for atorvastatin (54% and 24%, respectively), 2OH-atorvastatin (53% and 22%, respectively), and 4OH-atorvastatin (44% and 52%, respectively). Pharmacokinetics of atorvastatin were not significantly affected by coadministration with lamotrigine. **Evidence:** External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** PubMed
	Atorvastatin + Phenytoin **Findings:** Phenytoin significantly reduced atorvastatin bioavailability in healthy volunteers: atorvastatin AUC and Cmax fell by 54% and 24%, respectively, when coadministered with phenytoin versus atorvastatin alone. In this open-label pharmacokinetic study, volunteers received atorvastatin 40 mg/day before adding phenytoin, and the authors concluded that atorvastatin coadministered with phenytoin may require atorvastatin dose adjustment to maintain exposure. **Evidence:** External evidence (PubMed); ClinicalTrials.gov

POSITIVE IMPACT	NEGATIVE IMPACT
	Hyperlink: PubMed

BUPROPION

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Bupropion + Carbamazepine** **Findings:** Carbamazepine may induce bupropion metabolism and may decrease bupropion exposure when the two are used concomitantly. The DailyMed label states this has not been systematically studied for carbamazepine, but recommends that a bupropion dose increase may be necessary based on clinical response and should not exceed the maximum recommended dose. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed **Bupropion + Fluoxetine** **Findings:** Bupropion can increase exposure to CYP2D6 substrates, including fluoxetine, because bupropion and its metabolites inhibit CYP2D6. The DailyMed label recommends that when CYP2D6 substrates are used concomitantly with bupropion, it may be necessary to decrease the substrate dose, particularly for drugs with a narrow therapeutic index. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed **Bupropion + Haloperidol** **Findings:** Bupropion inhibits CYP2D6 and can increase concentrations of CYP2D6-metabolized antipsychotics, including haloperidol. The DailyMed label recommends considering dose reduction of these CYP2D6 substrates when used with bupropion, particularly for drugs with a narrow therapeutic index. **Evidence:** DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed **Bupropion + Nortriptyline** **Findings:** Bupropion inhibits CYP2D6 and can increase exposure to CYP2D6 substrates, including nortriptyline. The DailyMed label recommends considering dose reduction of nortriptyline when it is used concomitantly with bupropion, particularly for drugs with a narrow therapeutic index. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed **Bupropion + Paroxetine** **Findings:** Bupropion can increase exposure to CYP2D6 substrates, including paroxetine, because bupropion and its metabolites inhibit CYP2D6. The label states that when CYP2D6 substrates are used concomitantly with bupropion, it may be necessary to decrease the CYP2D6-substrate dose, particularly for drugs with a narrow therapeutic index. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed **Bupropion + Phenytoin** **Findings:** Phenytoin may induce the metabolism of bupropion and may decrease bupropion exposure, according to the bupropion DailyMed Drug Interactions section. The label notes this has not been systematically studied for carbamazepine, phenobarbital, and phenytoin, and recommends that if bupropion is used concomitantly with a CYP inducer, a bupropion dose increase may be necessary based on clinical response, without exceeding the maximum recommended dose. **Evidence:** DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** DailyMed **Bupropion + Sertraline** **Findings:** Bupropion inhibits CYP2D6 and can increase exposure to CYP2D6 substrates, including sertraline. The DailyMed label states that when sertraline is used concomitantly with bupropion, it may be necessary to decrease the dose of the CYP2D6 substrate, particularly for drugs with a narrow therapeutic index. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed **Bupropion + Venlafaxine** **Findings:** Bupropion inhibits CYP2D6 and can increase exposure to CYP2D6 substrates, including the antidepressant venlafaxine. The DailyMed label states that

POSITIVE IMPACT	NEGATIVE IMPACT
	when venlafaxine is used concomitantly with bupropion, dose reduction of the CYP2D6 substrate may be necessary, particularly for drugs with a narrow therapeutic index. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed

CARBAMAZEPINE

POSITIVE IMPACT	NEGATIVE IMPACT
Carbamazepine + Perampanel **Findings:** Perampanel's DailyMed label states: "Contraceptives: 12 mg once daily may decrease the effectiveness of hormonal contraceptives containing levonorgestrel (7.1) Moderate and Strong CYP3A4 Inducers (including carbamazepine, oxcarbazepine, and phenytoin): increase clearance of perampanel and decrease perampanel plasma concentrations." **Evidence:** DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** DailyMed	**Carbamazepine + Aripiprazole** **Findings:** Aripiprazole's DailyMed label states: "Strong CYP3A4 Inducers (e.g., carbamazepine, rifampin) Concomitant use of aripiprazole and carbamazepine decreased the exposure of aripiprazole compared to the use of aripiprazole alone [see Clinical Pharmacology" **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed **Carbamazepine + Atorvastatin** **Findings:** Chronic atorvastatin attenuated carbamazepine's anticonvulsant effect in a mouse maximal electroshock seizure model, increasing carbamazepine's ED50 against maximal electroconvulsions. The study evaluated acute and 7-day statin treatment with antiepileptic drugs in mice and also assessed adverse-effect potential and antiepileptic-drug brain concentrations, but the provided evidence does not state a clinical recommendation for this pair. **Evidence:** External evidence (PubMed) **Hyperlink:** PubMed **Carbamazepine + Bupropion** **Findings:** Carbamazepine may induce bupropion metabolism and may decrease bupropion exposure when the two are used concomitantly. The DailyMed label states this has not been systematically studied for carbamazepine, but recommends that a bupropion dose increase may be necessary based on clinical response and should not exceed the maximum recommended dose. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed **Carbamazepine + Escitalopram** **Findings:** The label states that coadministration with carbamazepine might increase the clearance of escitalopram, based on carbamazepine's enzyme-inducing properties, although a study of racemic citalopram 40 mg/day with carbamazepine titrated to 400 mg/day found no significant effect on carbamazepine pharmacokinetics and unaffected trough citalopram levels. The label recommends considering this possible increased escitalopram clearance if escitalopram and carbamazepine are coadministered. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed **Carbamazepine + Fluoxetine** **Findings:** Fluoxetine's label states there is a potential for elevated carbamazepine levels and clinical anticonvulsant toxicity when anticonvulsants such as carbamazepine are used with fluoxetine. The Drug Interactions section identifies fluoxetine as a potent CYP2D6 inhibitor and advises caution with centrally acting drugs, but the retrieved text does not give a carbamazepine-specific dose adjustment or quantitative exposure change. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed **Carbamazepine + Lamotrigine** **Findings:** Carbamazepine decreases lamotrigine concentrations by approximately 40% according to the lamotrigine DailyMed Drug Interactions section. The label states this occurs with addition of carbamazepine, notes decreased lamotrigine via induced glucuronidation, and says carbamazepine may increase carbamazepine epoxide levels. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed **Carbamazepine + Nortriptyline** **Findings:** During concomitant carbamazepine 500-600 mg/day, serum nortriptyline concentrations fell from a prior mean of 355 +/- 49 nM on nortriptyline 75 mg/day to 140 and 134 nM, while depression persisted. In this case report of a 73-year-old

POSITIVE IMPACT	NEGATIVE IMPACT
	woman, increasing nortriptyline to 150 mg/day during continued carbamazepine raised the mean serum nortriptyline concentration to 402 +/- 69 nM and the depression disappeared; the authors concluded that carbamazepine induces nortriptyline metabolism. **Evidence:** External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** PubMed
	Carbamazepine + Olanzapine **Findings:** Carbamazepine causes an approximately 50% increase in olanzapine clearance, and higher carbamazepine doses may cause an even greater increase. The DailyMed olanzapine label attributes this to carbamazepine being a potent inducer of CYP1A2 activity. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed
	Carbamazepine + Oxcarbazepine **Findings:** Carbamazepine coadministration may decrease blood levels of oxcarbazepine's active metabolite MHD, and the label recommends monitoring MHD plasma levels during oxcarbazepine titration. The DailyMed Drug Interactions section identifies carbamazepine as a strong CYP3A4/UGT inducer and states that oxcarbazepine dosage adjustment may be required after initiation, dosage modification, or discontinuation of such inducers. **Evidence:** DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** DailyMed
	Carbamazepine + Quetiapine **Findings:** Quetiapine's DailyMed label states: "prototype CYP3A4 inhibitors (e.g., ketoconazole, itraconazole, indinavir, ritonavir, nefazodone, etc.) and decreased by the prototype CYP3A4 inducers (e.g., phenytoin, carbamazepine, rifampin, avasimibe, St. John's wort etc.)." **Evidence:** DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed
	Carbamazepine + Sertraline **Findings:** General web evidence states that carbamazepine co-administered with sertraline accelerates hepatic breakdown of sertraline, reducing sertraline plasma concentrations; the induction effect is described as developing over 1–3 weeks and persisting for weeks after carbamazepine discontinuation. The source identifies the main clinical risk as reduced sertraline effectiveness, with possible worsening depression, anxiety, relapse, or treatment failure, and recommends close symptom monitoring. It also states that clinicians may need to increase the sertraline dose, typically by 25–50%, to maintain therapeutic effectiveness. **Evidence:** External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** General web search (Tavily)
	Carbamazepine + Valproic Acid **Findings:** Carbamazepine can increase valproate clearance; the label states that hepatic enzyme-inducing drugs such as carbamazepine can increase valproate clearance and that carbamazepine can double the clearance of valproate. The DailyMed drug-interactions text recommends increased monitoring of valproate and concomitant drug concentrations, with dosage adjustment indicated when enzyme-inducing drugs are introduced or withdrawn. **Evidence:** DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** DailyMed

CITALOPRAM

POSITIVE IMPACT	NEGATIVE IMPACT
Citalopram + Methylphenidate **Findings:** In ClinicalTrials.gov trial NCT00794040, add-on citalopram following optimized methylphenidate produced a statistically significant difference in the percentage of youth rated "much improved" or "completely recovered" from irritability severity compared with add-on placebo following optimized methylphenidate (p=0.006). The completed phase 2 trial studied youths with severe mood dysregulation/ADHD-related conditions after optimized methylphenidate treatment. **Evidence:** ClinicalTrials.gov **Hyperlink:** ClinicalTrials.gov	**Citalopram + Escitalopram** **Findings:** Combined citalopram and escitalopram have additive serotonergic activity, increasing the risk of serotonin syndrome, activation, sleep disturbance, and agitation. This pharmacodynamic rule attributes the risk to both agents' serotonergic effects when used together. **Evidence:** Pharmacodynamic interaction reference; DailyMed label (Drug Interactions section)
	Citalopram + Fluoxetine **Findings:** Combined citalopram and fluoxetine have additive serotonergic activity, which increases the risk of serotonin syndrome, activation, sleep disturbance, and

POSITIVE IMPACT	NEGATIVE IMPACT
	agitation. This pharmacodynamic interaction reference attributes the concern to combining serotonergic agents. **Evidence:** Pharmacodynamic interaction reference
	Citalopram + Lithium **Findings:** Concomitant use of citalopram with lithium increases the risk of serotonin syndrome because lithium is included among other serotonergic drugs in the citalopram label. The label recommends monitoring for signs and symptoms of serotonin syndrome, particularly during citalopram initiation and dosage increases, and considering discontinuation of citalopram and/or the concomitant serotonergic drug if serotonin syndrome occurs. **Evidence:** DailyMed label (Drug Interactions section); External evidence (PubMed) **Hyperlink:** DailyMed
	Citalopram + Paroxetine **Findings:** Combined citalopram and paroxetine have additive serotonergic activity, increasing the risk of serotonin syndrome, activation, sleep disturbance, and agitation. This pharmacodynamic rule attributes the risk to the serotonergic properties of both agents when used together. **Evidence:** Pharmacodynamic interaction reference; General web search (Tavily)
	Citalopram + Sertraline **Findings:** Combined use of citalopram and sertraline has additive serotonergic activity and is associated with increased risk of serotonin syndrome, activation, sleep disturbance, and agitation. This pharmacodynamic interaction reference attributes the risk to overlapping serotonergic effects from both agents. **Evidence:** Pharmacodynamic interaction reference; External evidence (PubMed); General web search (Tavily)

CLOBAZAM

POSITIVE IMPACT	NEGATIVE IMPACT
Clobazam + Quetiapine **Findings:** Discontinuation of clobazam and quetiapine was associated with clinical improvement in a 69-year-old man who developed MRI findings suggestive of central pontine myelinolysis despite normal serum sodium. This PubMed case report describes medication-induced CPM as a consideration for patients taking this combination who present with neurological symptoms, while preserving that the report presents a suspected association rather than a proven causal interaction. **Evidence:** External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** PubMed	No pairs currently fall in this category.

DULOXETINE

POSITIVE IMPACT	NEGATIVE IMPACT

Duloxetine + Methylphenidate

Findings: In this lactation case series, a mother taking duloxetine 90 mg daily and methylphenidate 36 mg daily while breastfeeding had low measured breast-milk concentrations, with estimated relative infant doses of 0.3% for duloxetine and 0.2% for methylphenidate at 4 weeks postpartum. The authors concluded that exposure to duloxetine and methylphenidate through breast milk appears minimal and that these drugs are likely compatible with lactation, while noting that large cohort studies are needed to evaluate long-term effects in exposed infants.
Evidence: External evidence (PubMed); ClinicalTrials.gov
Hyperlink: PubMed

Duloxetine + Olanzapine

Findings: The trial group, scored at 13, showed significantly less occurrence and severity of nausea and vomiting than the control group, scored at 38 ($P < 0.05$). Duloxetine combined with low-dose olanzapine can effectively reduce the occurrence and severity of duloxetine-related nausea and vomiting.
Evidence: External evidence (PubMed); ClinicalTrials.gov
Hyperlink: PubMed

Duloxetine + Valproic Acid

Findings: Duloxetine ameliorated valproic acid-induced hyperactivity, anxiety-like behavior, and social interaction deficits in a zebrafish model of ASD-like phenotypes. In this experimental study, embryonic valproic acid exposure produced persistent ASD-like behavioral and biochemical changes, and duloxetine was evaluated for rapid and persistent neuroprotective effects against those VPA-triggered deficits.
Evidence: External evidence (PubMed); ClinicalTrials.gov
Hyperlink: PubMed

Duloxetine + Fluoxetine

Findings: Fluoxetine is expected to have effects similar to other potent CYP2D6 inhibitors that increase duloxetine exposure; the label reports that duloxetine 40 mg once daily with paroxetine 20 mg once daily increased duloxetine AUC by about 60%, with greater inhibition expected at higher paroxetine doses. The DailyMed Drug Interactions section identifies fluoxetine as an example of a potent CYP2D6 inhibitor and links this expected increase in duloxetine exposure to the label's warnings/precautions.
Evidence: DailyMed label (Drug Interactions section)
Hyperlink: DailyMed

Duloxetine + Levetiracetam

Findings: Levetiracetam plus duloxetine produced a marked synergistic antinociceptive interaction in diabetic mice, with about an 18-fold reduction in the doses of both drugs in the combination. This PubMed-indexed mouse diabetic painful neuropathy study coadministered fixed-dose fractions of each single-drug ED50 and used isobolographic analysis; the authors concluded that levetiracetam combinations with duloxetine and other agents reduced nociception and might be of benefit for diabetic patients, warranting further clinical trials.
Evidence: External evidence (PubMed); ClinicalTrials.gov
Hyperlink: PubMed

Duloxetine + Lithium

Findings: Water diuresis was induced in the Li-NDI rats, and duloxetine treatment reduced polyuria while increasing urine osmolality. Duloxetine treatment prevented the decrease in total AQP2, AQP2 phosphorylation at
Evidence: External evidence (PubMed); ClinicalTrials.gov
Hyperlink: PubMed

Duloxetine + Paroxetine

Findings: Duloxetine's DailyMed label states: "7.2 Inhibitors of CYP2D6 Concomitant use of duloxetine (40 mg once daily) with paroxetine (20 mg once daily) increased the concentration of duloxetine AUC by about 60%, and greater degrees of inhibition are expected with higher doses of paroxetine."
Evidence: DailyMed label (Drug Interactions section)
Hyperlink: DailyMed

ESCITALOPRAM

POSITIVE IMPACT	NEGATIVE IMPACT

Escitalopram + Oxcarbazepine

Findings: In a 108-patient comparative study of epilepsy with depressive disorder, oxcarbazepine combined with escitalopram showed no remarkable difference in total efficacy versus lamotrigine combined with escitalopram after six months. Seizure number and duration, EEG epileptic discharge, and quality of life improved after treatment in both combination groups with no marked between-group difference, while the available abstract text indicates HAMD and MADRS scores were significantly lower in the lamotrigine-plus-escitalopram group.
Evidence: External evidence (PubMed); ClinicalTrials.gov
Hyperlink: PubMed

Escitalopram + Carbamazepine

Findings: The label states that coadministration with carbamazepine might increase the clearance of escitalopram, based on carbamazepine's enzyme-inducing properties, although a study of racemic citalopram 40 mg/day with carbamazepine titrated to 400 mg/day found no significant effect on carbamazepine pharmacokinetics and unaffected trough citalopram levels. The label recommends considering this possible increased escitalopram clearance if escitalopram and carbamazepine are coadministered.
Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov
Hyperlink: DailyMed

Escitalopram + Citalopram

Findings: Combined citalopram and escitalopram have additive serotonergic activity, increasing the risk of serotonin syndrome, activation, sleep disturbance, and agitation. This pharmacodynamic rule attributes the risk to both agents' serotonergic effects when used together.
Evidence: Pharmacodynamic interaction reference; DailyMed label (Drug Interactions section)

Escitalopram + Fluoxetine

Findings: Combined escitalopram and fluoxetine have additive serotonergic activity, with increased risk of serotonin syndrome, activation, sleep disturbance, and agitation. This pharmacodynamic interaction reference attributes the risk to combining serotonergic agents and supports clinical concern for additive serotonergic adverse effects with this pair.
Evidence: Pharmacodynamic interaction reference

Escitalopram + Lithium

Findings: Escitalopram's DailyMed label states: "Clinical Impact: Concomitant use of escitalopram and other serotonergic drugs (including other SSRIs, SNRIs, SNRIs, triptans, tricycle antidepressant, opioids, lithium, buspirone, amphetamines, tryptophan and St. John's Wort) increase the risk of serotonin syndrome."

POSITIVE IMPACT	NEGATIVE IMPACT
	Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed
	Escitalopram + Paroxetine **Findings:** Combining escitalopram and paroxetine has additive serotonergic activity and increases the risk of serotonin syndrome, activation, sleep disturbance, and agitation. This pharmacodynamic rule is based on both drugs being serotonergic agents; the clinically relevant implication is to monitor for serotonergic toxicity and related activation or sleep/agitation symptoms when they are used together. **Evidence:** Pharmacodynamic interaction reference
	Escitalopram + Sertraline **Findings:** Combined escitalopram and sertraline have additive serotonergic activity, increasing the risk of serotonin syndrome as well as activation, sleep disturbance, and agitation. This pharmacodynamic interaction reference attributes the risk to both drugs acting as serotonergic agents when used together. **Evidence:** Pharmacodynamic interaction reference; External evidence (PubMed); ClinicalTrials.gov

ETHOSUXIMIDE

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Ethosuximide + Phenytoin** **Findings:** Ethosuximide may elevate phenytoin serum levels when used with concurrently administered antiepileptic drugs. The DailyMed Drug Interactions section says periodic serum level determinations of these drugs may be necessary. **Evidence:** DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed
	Ethosuximide + Valproic Acid **Findings:** Valproic acid has been reported to both increase and decrease ethosuximide levels when used concurrently. The DailyMed ethosuximide label recommends periodic serum level determinations for concurrently administered antiepileptic drugs because ethosuximide may interact with them. **Evidence:** DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** DailyMed

FLUOXETINE

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Fluoxetine + Aripiprazole** **Findings:** Fluoxetine, listed as a strong CYP2D6 inhibitor in the aripiprazole label, increases aripiprazole exposure when used concomitantly compared with aripiprazole alone. The label recommends reducing the aripiprazole dosage when it is administered with a strong CYP2D6 inhibitor such as fluoxetine. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed
	Fluoxetine + Bupropion **Findings:** Bupropion can increase exposure to CYP2D6 substrates, including fluoxetine, because bupropion and its metabolites inhibit CYP2D6. The DailyMed label recommends that when CYP2D6 substrates are used concomitantly with bupropion, it may be necessary to decrease the substrate dose, particularly for drugs with a narrow therapeutic index. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed
	Fluoxetine + Carbamazepine **Findings:** Fluoxetine's label states there is a potential for elevated carbamazepine levels and clinical anticonvulsant toxicity when anticonvulsants such as carbamazepine are used with fluoxetine. The Drug Interactions section identifies fluoxetine as a potent CYP2D6 inhibitor and advises caution with centrally acting drugs, but the retrieved text does not give a carbamazepine-specific dose adjustment or quantitative exposure change. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed

POSITIVE IMPACT	NEGATIVE IMPACT

Fluoxetine + Citalopram

Findings: Combined citalopram and fluoxetine have additive serotonergic activity, which increases the risk of serotonin syndrome, activation, sleep disturbance, and agitation. This pharmacodynamic interaction reference attributes the concern to combining serotonergic agents.

Evidence: Pharmacodynamic interaction reference

Fluoxetine + Duloxetine

Findings: Fluoxetine is expected to have effects similar to other potent CYP2D6 inhibitors that increase duloxetine exposure; the label reports that duloxetine 40 mg once daily with paroxetine 20 mg once daily increased duloxetine AUC by about 60%, with greater inhibition expected at higher paroxetine doses. The DailyMed Drug Interactions section identifies fluoxetine as an example of a potent CYP2D6 inhibitor and links this expected increase in duloxetine exposure to the label's warnings/precautions.

Evidence: DailyMed label (Drug Interactions section)

Hyperlink: DailyMed

Fluoxetine + Escitalopram

Findings: Combined escitalopram and fluoxetine have additive serotonergic activity, with increased risk of serotonin syndrome, activation, sleep disturbance, and agitation. This pharmacodynamic interaction reference attributes the risk to combining serotonergic agents and supports clinical concern for additive serotonergic adverse effects with this pair.

Evidence: Pharmacodynamic interaction reference

Fluoxetine + Haloperidol

Findings: Fluoxetine has the potential to elevate haloperidol levels, according to the DailyMed Drug Interactions section. The label identifies fluoxetine as a potent CYP2D6 inhibitor and advises caution when it is used with other centrally acting drugs or drugs that prolong the QT interval.

Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); ClinicalTrials.gov

Hyperlink: DailyMed

Fluoxetine + Lithium

Findings: Concomitant use of fluoxetine with lithium increases the risk of serotonin syndrome, according to the fluoxetine DailyMed Drug Interactions section. The label advises monitoring for signs and symptoms of serotonin syndrome, particularly during treatment initiation and dosage increases, and considering discontinuation of fluoxetine and/or the concomitant serotonergic drug if serotonin syndrome occurs.

Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov

Hyperlink: DailyMed

Fluoxetine + Methylphenidate

Findings: Chronic oral methylphenidate plus fluoxetine did not amplify methylphenidate-associated microglial activation in this rat study. Three-week-old male Sprague Dawley rats received water, methylphenidate, fluoxetine, or the combination for four weeks; methylphenidate and the combination significantly increased [3H] PK11195 binding in several somatosensory regions, but the authors concluded that co-administration with fluoxetine did not amplify these effects.

Evidence: External evidence (PubMed)

Hyperlink: PubMed

Fluoxetine + Olanzapine

Findings: Fluoxetine's label does not state a specific pharmacokinetic or clinical consequence for olanzapine when the two are used together; it instructs readers to refer to the Symbyax package insert for drug-interaction information. The only pair-specific label statement in the retrieved text is that olanzapine may be used in combination with fluoxetine, without providing an outcome, mechanism, dose adjustment, or monitoring recommendation in this excerpt.

Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily)

Hyperlink: DailyMed

Fluoxetine + Paroxetine

Findings: Combined fluoxetine and paroxetine have additive serotonergic activity, increasing the risk of serotonin syndrome, activation, sleep disturbance, and agitation. This pharmacodynamic interaction reference attributes the risk to the serotonergic properties of both agents.

Evidence: Pharmacodynamic interaction reference; External evidence (PubMed)

Fluoxetine + Phenytoin

POSITIVE IMPACT	NEGATIVE IMPACT
	Findings: Fluoxetine's label states there is potential for elevated phenytoin levels and clinical anticonvulsant toxicity when anticonvulsants such as phenytoin are used with fluoxetine. This is a DailyMed Drug Interactions label statement; it identifies phenytoin as an anticonvulsant for which elevated levels and toxicity are a potential concern during coadministration. **Evidence:** DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** DailyMed
	Fluoxetine + Sertraline **Findings:** Combined fluoxetine and sertraline has additive serotonergic activity and increases the risk of serotonin syndrome, activation, sleep disturbance, and agitation. The pharmacodynamic rule attributes this risk to combining serotonergic agents, so the clinically relevant implication is increased serotonergic adverse-effect risk when these drugs are used together. **Evidence:** Pharmacodynamic interaction reference; External evidence (PubMed); General web search (Tavily)

HALOPERIDOL

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Haloperidol + Bupropion** **Findings:** Bupropion inhibits CYP2D6 and can increase concentrations of CYP2D6-metabolized antipsychotics, including haloperidol. The DailyMed label recommends considering dose reduction of these CYP2D6 substrates when used with bupropion, particularly for drugs with a narrow therapeutic index. **Evidence:** DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed
	Haloperidol + Fluoxetine **Findings:** Fluoxetine has the potential to elevate haloperidol levels, according to the DailyMed Drug Interactions section. The label identifies fluoxetine as a potent CYP2D6 inhibitor and advises caution when it is used with other centrally acting drugs or drugs that prolong the QT interval. **Evidence:** DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed
	Haloperidol + Methylphenidate **Findings:** A haloperidol challenge dose attenuated methylphenidate-associated increases in motor activity and impaired cognitive performance in rats previously treated long term with methylphenidate. In this animal study, methylphenidate was given for 13 days at 0.6, 2.5, or 10 mg/kg/day, and haloperidol 1 mg/kg i.p. was used as a challenge dose to assess dopamine D2-receptor sensitivity. **Evidence:** External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** PubMed
	Haloperidol + Phenytoin **Findings:** Phenytoin coadministration accelerates haloperidol hepatic metabolism, increasing haloperidol clearance and reducing its plasma concentrations; this induction effect typically develops over 1–2 weeks and can persist for several weeks after phenytoin discontinuation. The source states this may reduce haloperidol effectiveness and risk breakthrough or worsening psychiatric symptoms, and recommends close monitoring when phenytoin is started or stopped, with possible haloperidol dose increase of 50–100% when phenytoin is added. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** General web search (Tavily)
	Haloperidol + Venlafaxine **Findings:** Venlafaxine's DailyMed label states: "(5.2) Cimetidine: Caution in patients with pre-existing hypertension, in elderly patients and patients with hepatic dysfunction. (7.2) Haloperidol: Increase in haloperidol AUC and C max . (7.4) Ketoconazole: Increase in venlafaxine and O-desmethylvenlafaxine AUC and C max ." **Evidence:** DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** DailyMed

LAMOTRIGINE

POSITIVE IMPACT	NEGATIVE IMPACT
Lamotrigine + Oxcarbazepine **Findings:** Lamotrigine and oxcarbazepine did not significantly affect each other's pharmacokinetics at steady state, but adverse events occurred more frequently with the combination than with either drug plus placebo. In this randomized, single-blind study in 47 healthy male volunteers, lamotrigine 200 mg daily was coadministered with oxcarbazepine 600 mg twice daily; lamotrigine AUC(0–24) and Cmax and oxcarbazepine active metabolite MHD AUC(0–12) and Cmax were not significantly changed. **Evidence:** ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** General web search (Tavily)	**Lamotrigine + Atorvastatin** **Findings:** When atorvastatin was administered with lamotrigine compared with when atorvastatin was administered alone, atorvastatin AUC((0-τ)) was within bounds indicating no interaction, whereas C(max) was slightly higher(14%); AUC((0-τ)) and C(max) were 3% and 20% higher, respectively, for 2OH-atorvastatin and 25% and 21% higher, respectively, for 4OH-atorvastatin.When atorvastatin was administered with phenytoin compared with when atorvastatin was administered alone, reductions in AUC((0-τ)) and C(max) were observed for atorvastatin (54% and 24%, respectively), 2OH-atorvastatin (53% and 22%, respectively), and 4OH-atorvastatin (44% and 52%, respectively). Pharmacokinetics of atorvastatin were not significantly affected by coadministration with lamotrigine. **Evidence:** External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** PubMed
Lamotrigine + Paroxetine **Findings:** Further addition of paroxetine did not appear to provide additional benefit in non-responders already receiving lithium plus lamotrigine. In this randomized bipolar-depression treatment algorithm, patients first received lithium with lamotrigine or placebo for 8 weeks, then non-responders received paroxetine 20 mg for another 8 weeks; the source reports benefit from lamotrigine addition initially, but not from adding paroxetine to the lithium-plus-lamotrigine regimen. **Evidence:** External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** PubMed	**Lamotrigine + Carbamazepine** **Findings:** Carbamazepine decreases lamotrigine concentrations by approximately 40% according to the lamotrigine DailyMed Drug Interactions section. The label states this occurs with addition of carbamazepine, notes decreased lamotrigine via induced glucuronidation, and says carbamazepine may increase carbamazepine epoxide levels. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed
	Lamotrigine + Phenytoin **Findings:** Phenytoin decreases lamotrigine concentrations by approximately 40%. The DailyMed lamotrigine label identifies UGT enzymes as responsible for lamotrigine metabolism and lists phenytoin in the Drug Interactions section as decreasing lamotrigine concentration; no additional management recommendation for this specific pair is provided in the retrieved text. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed

LEVETIRACETAM

POSITIVE IMPACT	NEGATIVE IMPACT
Levetiracetam + Olanzapine **Findings:** There were no differences in the baseline characteristics between the case and control groups. Levetiracetam combined with olanzapine may benefit BD patients unresponsive to olanzapine alone. **Evidence:** External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** PubMed	**Levetiracetam + Duloxetine** **Findings:** Levetiracetam plus duloxetine produced a marked synergistic antinociceptive interaction in diabetic mice, with about an 18-fold reduction in the doses of both drugs in the combination. This PubMed-indexed mouse diabetic painful neuropathy study coadministered fixed-dose fractions of each single-drug ED50 and used isobolographic analysis; the authors concluded that levetiracetam combinations with duloxetine and other agents reduced nociception and might be of benefit for diabetic patients, warranting further clinical trials. **Evidence:** External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** PubMed

LITHIUM

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Lithium + Citalopram** **Findings:** Concomitant use of citalopram with lithium increases the risk of serotonin syndrome because lithium is included among other serotonergic drugs in the citalopram label. The label recommends monitoring for signs and symptoms of serotonin syndrome, particularly during citalopram initiation and dosage increases, and considering discontinuation of citalopram and/or the concomitant serotonergic drug if serotonin syndrome occurs. **Evidence:** DailyMed label (Drug Interactions section); External evidence (PubMed) **Hyperlink:** DailyMed
	Lithium + Duloxetine **Findings:** Water diuresis was induced in the Li-NDI rats, and duloxetine treatment reduced polyuria while increasing urine osmolality. Duloxetine treatment prevented the decrease in total AQP2, AQP2 phosphorylation at **Evidence:** External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** PubMed
	Lithium + Escitalopram **Findings:** Escitalopram's DailyMed label states: "Clinical Impact: Concomitant use of escitalopram and other serotonergic drugs (including other SSRIs, SNRIs, SNRIs, triptans, tricycle antidepressant, opioids, lithium, buspirone, amphetamines, tryptophan and St. John's Wort) increase the risk of serotonin syndrome."

POSITIVE IMPACT	NEGATIVE IMPACT
	Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed
	Lithium + Fluoxetine **Findings:** Concomitant use of fluoxetine with lithium increases the risk of serotonin syndrome, according to the fluoxetine DailyMed Drug Interactions section. The label advises monitoring for signs and symptoms of serotonin syndrome, particularly during treatment initiation and dosage increases, and considering discontinuation of fluoxetine and/or the concomitant serotonergic drug if serotonin syndrome occurs. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed
	Lithium + Methylphenidate **Findings:** In a mouse behavioral model, lithium pretreatment significantly prevented methylphenidate-induced hyperlocomotion and was more effective than valproate in preventing methylphenidate-induced mania-like behaviors in the hole-board test. The study administered methylphenidate 10 mg/kg intraperitoneally after lithium 50 mg/kg intraperitoneally and evaluated open-field and hole-board behaviors related to mania-like locomotion, goal-directed action, and risk-taking. **Evidence:** External evidence (PubMed) **Hyperlink:** PubMed
	Lithium + Venlafaxine **Findings:** Venlafaxine's label states that a single 600 mg oral dose of lithium did not affect steady-state venlafaxine pharmacokinetics at 150 mg/day in 12 healthy male subjects, and O-desmethylvenlafaxine was also unaffected. The label also states that venlafaxine had no effect on lithium pharmacokinetics, but separately warns that serotonergic drugs including lithium have a potential for serotonin syndrome and advises careful patient observation. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed

METHYLPHENIDATE

POSITIVE IMPACT	NEGATIVE IMPACT
Methylphenidate + Citalopram **Findings:** In ClinicalTrials.gov trial NCT00794040, add-on citalopram following optimized methylphenidate produced a statistically significant difference in the percentage of youth rated "much improved" or "completely recovered" from irritability severity compared with add-on placebo following optimized methylphenidate (p=0.006). The completed phase 2 trial studied youths with severe mood dysregulation/ADHD-related conditions after optimized methylphenidate treatment. **Evidence:** ClinicalTrials.gov **Hyperlink:** ClinicalTrials.gov	**Methylphenidate + Fluoxetine** **Findings:** Chronic oral methylphenidate plus fluoxetine did not amplify methylphenidate-associated microglial activation in this rat study. Three-week-old male Sprague Dawley rats received water, methylphenidate, fluoxetine, or the combination for four weeks; methylphenidate and the combination significantly increased [3H] PK11195 binding in several somatosensory regions, but the authors concluded that co-administration with fluoxetine did not amplify these effects. **Evidence:** External evidence (PubMed) **Hyperlink:** PubMed
Methylphenidate + Duloxetine **Findings:** In this lactation case series, a mother taking duloxetine 90 mg daily and methylphenidate 36 mg daily while breastfeeding had low measured breast-milk concentrations, with estimated relative infant doses of 0.3% for duloxetine and 0.2% for methylphenidate at 4 weeks postpartum. The authors concluded that exposure to duloxetine and methylphenidate through breast milk appears minimal and that these drugs are likely compatible with lactation, while noting that large cohort studies are needed to evaluate long-term effects in exposed infants. **Evidence:** External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** PubMed	**Methylphenidate + Haloperidol** **Findings:** A haloperidol challenge dose attenuated methylphenidate-associated increases in motor activity and impaired cognitive performance in rats previously treated long term with methylphenidate. In this animal study, methylphenidate was given for 13 days at 0.6, 2.5, or 10 mg/kg/day, and haloperidol 1 mg/kg i.p. was used as a challenge dose to assess dopamine D2-receptor sensitivity. **Evidence:** External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** PubMed
Methylphenidate + Quetiapine **Findings:** Adding quetiapine to ongoing OROS methylphenidate was associated with significant improvement in ADHD symptoms, aggression, and global functioning in adolescents who had not responded sufficiently to methylphenidate alone; at the end of combination treatment, 42% met all criteria for clinically significant improvement and 79% showed minimal aggression. In this prospective open-label study of participants aged 12–16 years, methylphenidate was titrated to 54 mg/day before 9 weeks of combined methylphenidate-quetiapine treatment; mild transient sedation occurred in about half of cases, and prior methylphenidate-associated weight loss was offset by weight gain during combination treatment. **Evidence:** External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** PubMed	**Methylphenidate + Lithium** **Findings:** In a mouse behavioral model, lithium pretreatment significantly prevented methylphenidate-induced hyperlocomotion and was more effective than valproate in preventing methylphenidate-induced mania-like behaviors in the hole-board test. The study administered methylphenidate 10 mg/kg intraperitoneally after lithium 50 mg/kg intraperitoneally and evaluated open-field and hole-board behaviors related to mania-like locomotion, goal-directed action, and risk-taking. **Evidence:** External evidence (PubMed) **Hyperlink:** PubMed
Methylphenidate + Valproic Acid	**Methylphenidate + Phenytoin** **Findings:** Methylphenidate may increase phenytoin serum levels according to the phenytoin DailyMed Drug Interactions table, where methylphenidate is listed among "other" interacting agents under drugs that may increase phenytoin concentrations. The excerpt does not provide an effect size or a specific management recommendation for methylphenidate beyond the label's general goal of achieving optimal clinical outcome.

POSITIVE IMPACT	NEGATIVE IMPACT
Findings: Methylphenidate or atomoxetine for 2 weeks also improved the prenatal VPA-induced decrease in dendritic spine density in the prefrontal cortex. The effects of these drugs on behavior **Evidence:** External evidence (PubMed) **Hyperlink:** PubMed	**Evidence:** DailyMed label (Drug Interactions section); External evidence (PubMed); General web search (Tavily) **Hyperlink:** DailyMed

NORTRIPTYLINE

POSITIVE IMPACT	NEGATIVE IMPACT
Nortriptyline + Atorvastatin **Findings:** Atorvastatin added to nortriptyline significantly reduced migraine attack frequency versus placebo plus nortriptyline after 24 weeks (P = 0.004), improved quality of life at follow-up intervals including week 14 (P = 0.001) and week 24, and did not improve headache intensity. This was a prospective, triple-blind randomized controlled trial in 68 adult migraine patients, with constipation reported as the most common adverse effect in the atorvastatin combination arm. **Evidence:** External evidence (PubMed) **Hyperlink:** PubMed	**Nortriptyline + Bupropion** **Findings:** Bupropion inhibits CYP2D6 and can increase exposure to CYP2D6 substrates, including nortriptyline. The DailyMed label recommends considering dose reduction of nortriptyline when it is used concomitantly with bupropion, particularly for drugs with a narrow therapeutic index. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed
Nortriptyline + Valproic Acid **Findings:** Valproic acid's label states that dosage adjustment of amitriptyline/nortriptyline may be necessary when used concomitantly with valproic acid. The excerpt describes this as part of valproate's ability to affect the pharmacokinetics of other drugs by inhibiting metabolism or displacing protein binding, and recommends monitoring concomitant drug concentrations when enzyme-modifying drugs are introduced or withdrawn. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed	**Nortriptyline + Carbamazepine** **Findings:** During concomitant carbamazepine 500-600 mg/day, serum nortriptyline concentrations fell from a prior mean of 355 +/- 49 nM on nortriptyline 75 mg/day to 140 and 134 nM, while depression persisted. In this case report of a 73-year-old woman, increasing nortriptyline to 150 mg/day during continued carbamazepine raised the mean serum nortriptyline concentration to 402 +/- 69 nM and the depression disappeared; the authors concluded that carbamazepine induces nortriptyline metabolism. **Evidence:** External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** PubMed

OLANZAPINE

POSITIVE IMPACT	NEGATIVE IMPACT
Olanzapine + Atomoxetine **Findings:** Open-label treatment with atomoxetine plus olanzapine for 10 weeks was associated with statistically significant improvement in ADHD symptoms and overt aggression in children and adolescents with ADHD and comorbid disruptive behavior disorder. In 11 patients ages 10–18, symptoms fell by 33% on the ADHD-RS-I and MOAS; 73% responded for ADHD and 55% for aggression, and both medications were generally well tolerated, although olanzapine was associated with significant weight gain averaging 3.9 kg. **Evidence:** External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** PubMed	**Olanzapine + Carbamazepine** **Findings:** Carbamazepine causes an approximately 50% increase in olanzapine clearance, and higher carbamazepine doses may cause an even greater increase. The DailyMed olanzapine label attributes this to carbamazepine being a potent inducer of CYP1A2 activity. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed
Olanzapine + Duloxetine **Findings:** The trial group, scored at 13, showed significantly less occurrence and severity of nausea and vomiting than the control group, scored at 38 (P < 0.05). Duloxetine combined with low-dose olanzapine can effectively reduce the occurrence and severity of duloxetine-related nausea and vomiting. **Evidence:** External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** PubMed	**Olanzapine + Fluoxetine** **Findings:** Fluoxetine's label does not state a specific pharmacokinetic or clinical consequence for olanzapine when the two are used together; it instructs readers to refer to the Symbyax package insert for drug-interaction information. The only pair-specific label statement in the retrieved text is that olanzapine may be used in combination with fluoxetine, without providing an outcome, mechanism, dose adjustment, or monitoring recommendation in this excerpt. **Evidence:** DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed
Olanzapine + Levetiracetam **Findings:** There were no differences in the baseline characteristics between the case and control groups. Levetiracetam combined with olanzapine may benefit BD patients unresponsive to olanzapine alone. **Evidence:** External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** PubMed	

OXCARBAZEPINE

POSITIVE IMPACT	NEGATIVE IMPACT
Oxcarbazepine + Escitalopram **Findings:** In a 108-patient comparative study of epilepsy with depressive disorder, oxcarbazepine combined with escitalopram showed no remarkable difference in total efficacy versus lamotrigine combined with escitalopram after six months. Seizure number and duration, EEG epileptic discharge, and quality of life improved after treatment in both combination groups with no marked between-group difference, while the available abstract text indicates HAMD and MADRS scores were significantly lower in the lamotrigine-plus-escitalopram group. **Evidence:** External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** PubMed	**Oxcarbazepine + Carbamazepine** **Findings:** Carbamazepine coadministration may decrease blood levels of oxcarbazepine's active metabolite MHD, and the label recommends monitoring MHD plasma levels during oxcarbazepine titration. The DailyMed Drug Interactions section identifies carbamazepine as a strong CYP3A4/UGT inducer and states that oxcarbazepine dosage adjustment may be required after initiation, dosage modification, or discontinuation of such inducers. **Evidence:** DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** DailyMed

POSITIVE IMPACT	NEGATIVE IMPACT
Oxcarbazepine + Lamotrigine **Findings:** Lamotrigine and oxcarbazepine did not significantly affect each other's pharmacokinetics at steady state, but adverse events occurred more frequently with the combination than with either drug plus placebo. In this randomized, single-blind study in 47 healthy male volunteers, lamotrigine 200 mg daily was coadministered with oxcarbazepine 600 mg twice daily; lamotrigine AUC(0–24) and Cmax and oxcarbazepine active metabolite MHD AUC(0–12) and Cmax were not significantly changed. **Evidence:** ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** General web search (Tavily)	**Oxcarbazepine + Phenytoin** **Findings:** Oxcarbazepine and phenytoin affect each other's monitored drug levels when coadministered: the label states that phenytoin may decrease blood levels of oxcarbazepine's active metabolite MHD, and that oxcarbazepine titration or dosage changes warrant monitoring phenytoin plasma levels. The DailyMed label recommends monitoring MHD during concurrent use with phenytoin and says Oxtellar XR dosage adjustment may be required; it also recommends monitoring phenytoin during Oxtellar XR titration or dosage modification because a phenytoin dose decrease may be required. **Evidence:** DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed
Oxcarbazepine + Perampanel **Findings:** Perampanel's DailyMed label states: "daily may decrease the effectiveness of hormonal contraceptives containing levonorgestrel (7.1) Moderate and Strong CYP3A4 Inducers (including carbamazepine, oxcarbazepine, and phenytoin): increase clearance of perampanel and decrease perampanel plasma concentrations." **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed	

PAROXETINE

POSITIVE IMPACT	NEGATIVE IMPACT
Paroxetine + Lamotrigine **Findings:** Further addition of paroxetine did not appear to provide additional benefit in non-responders already receiving lithium plus lamotrigine. In this randomized bipolar-depression treatment algorithm, patients first received lithium with lamotrigine or placebo for 8 weeks, then non-responders received paroxetine 20 mg for another 8 weeks; the source reports benefit from lamotrigine addition initially, but not from adding paroxetine to the lithium-plus-lamotrigine regimen. **Evidence:** External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** PubMed	**Paroxetine + Aripiprazole** **Findings:** Paroxetine is listed as a strong CYP2D6 inhibitor whose concomitant use with aripiprazole increased aripiprazole exposure compared with aripiprazole alone. The DailyMed label recommends reducing the aripiprazole dosage when it is administered with a strong CYP2D6 inhibitor such as paroxetine. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed
	Paroxetine + Atomoxetine **Findings:** Paroxetine may increase atomoxetine exposure because paroxetine is a CYP2D6 inhibitor and atomoxetine is listed as a CYP2D6 substrate in the label's Drug Interactions section. The label recommends decreasing the dosage of the CYP2D6 substrate if needed during concomitant paroxetine use, and notes that an increase in the substrate dose may be needed if paroxetine is discontinued. **Evidence:** DailyMed label (Drug Interactions section); External evidence (PubMed); General web search (Tavily) **Hyperlink:** DailyMed
	Paroxetine + Bupropion **Findings:** Bupropion can increase exposure to CYP2D6 substrates, including paroxetine, because bupropion and its metabolites inhibit CYP2D6. The label states that when CYP2D6 substrates are used concomitantly with bupropion, it may be necessary to decrease the CYP2D6-substrate dose, particularly for drugs with a narrow therapeutic index. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed
	Paroxetine + Citalopram **Findings:** Combined citalopram and paroxetine have additive serotonergic activity, increasing the risk of serotonin syndrome, activation, sleep disturbance, and agitation. This pharmacodynamic rule attributes the risk to the serotonergic properties of both agents when used together. **Evidence:** Pharmacodynamic interaction reference; General web search (Tavily)
	Paroxetine + Duloxetine **Findings:** Duloxetine's DailyMed label states: "7.2 Inhibitors of CYP2D6 Concomitant use of duloxetine (40 mg once daily) with paroxetine (20 mg once daily) increased the concentration of duloxetine AUC by about 60%, and greater degrees of inhibition are expected with higher doses of paroxetine." **Evidence:** DailyMed label (Drug Interactions section) **Hyperlink:** DailyMed
	Paroxetine + Escitalopram **Findings:** Combining escitalopram and paroxetine has additive serotonergic activity and increases the risk of serotonin syndrome, activation, sleep disturbance, and agitation. This pharmacodynamic rule is based on both drugs being serotonergic agents; the clinically relevant implication is to monitor for serotonergic toxicity and related activation or sleep/agitation symptoms when they are used together.

POSITIVE IMPACT	NEGATIVE IMPACT
	Evidence: Pharmacodynamic interaction reference **Paroxetine + Fluoxetine** **Findings:** Combined fluoxetine and paroxetine have additive serotonergic activity, increasing the risk of serotonin syndrome, activation, sleep disturbance, and agitation. This pharmacodynamic interaction reference attributes the risk to the serotonergic properties of both agents. **Evidence:** Pharmacodynamic interaction reference; External evidence (PubMed) **Paroxetine + Sertraline** **Findings:** Paroxetine and sertraline have additive serotonergic activity when combined, increasing the risk of serotonin syndrome as well as activation, sleep disturbance, and agitation. This pharmacodynamic interaction reference attributes the risk to both agents' serotonergic properties and identifies the clinically relevant consequence as increased serotonergic adverse-effect risk. **Evidence:** Pharmacodynamic interaction reference; General web search (Tavily)

PERAMPANEL

POSITIVE IMPACT	NEGATIVE IMPACT
Perampanel + Carbamazepine **Findings:** Perampanel's DailyMed label states: "Contraceptives: 12 mg once daily may decrease the effectiveness of hormonal contraceptives containing levonorgestrel (7.1) Moderate and Strong CYP3A4 Inducers (including carbamazepine, oxcarbazepine, and phenytoin): increase clearance of perampanel and decrease perampanel plasma concentrations." **Evidence:** DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** DailyMed **Perampanel + Oxcarbazepine** **Findings:** Perampanel's DailyMed label states: "daily may decrease the effectiveness of hormonal contraceptives containing levonorgestrel (7.1) Moderate and Strong CYP3A4 Inducers (including carbamazepine, oxcarbazepine, and phenytoin): increase clearance of perampanel and decrease perampanel plasma concentrations." **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed **Perampanel + Phenytoin** **Findings:** Phenytoin, as a moderate or strong CYP3A4 inducer, decreases perampanel plasma levels by approximately 50% to 67% when used concomitantly. The DailyMed label recommends increasing perampanel starting doses in the presence of such inducers and closely monitoring clinical response and tolerability when phenytoin is introduced or withdrawn, because perampanel dose adjustment may be necessary. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed	No pairs currently fall in this category.

PHENYTOIN

POSITIVE IMPACT	NEGATIVE IMPACT
Phenytoin + Perampanel **Findings:** Phenytoin, as a moderate or strong CYP3A4 inducer, decreases perampanel plasma levels by approximately 50% to 67% when used concomitantly. The DailyMed label recommends increasing perampanel starting doses in the presence of such inducers and closely monitoring clinical response and tolerability when phenytoin is introduced or withdrawn, because perampanel dose adjustment may be necessary. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed	**Phenytoin + Atorvastatin** **Findings:** Phenytoin significantly reduced atorvastatin bioavailability in healthy volunteers: atorvastatin AUC and Cmax fell by 54% and 24%, respectively, when coadministered with phenytoin versus atorvastatin alone. In this open-label pharmacokinetic study, volunteers received atorvastatin 40 mg/day before adding phenytoin, and the authors concluded that atorvastatin coadministered with phenytoin may require atorvastatin dose adjustment to maintain exposure. **Evidence:** External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** PubMed **Phenytoin + Bupropion** **Findings:** Phenytoin may induce the metabolism of bupropion and may decrease bupropion exposure, according to the bupropion DailyMed Drug Interactions section. The label notes this has not been systematically studied for carbamazepine, phenobarbital, and phenytoin, and recommends that if bupropion is used concomitantly with a CYP inducer, a bupropion dose increase may be necessary based on clinical response, without exceeding the maximum recommended dose. **Evidence:** DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** DailyMed

POSITIVE IMPACT	NEGATIVE IMPACT

Phenytoin + Ethosuximide

Findings: Ethosuximide may elevate phenytoin serum levels when used with concurrently administered antiepileptic drugs. The DailyMed Drug Interactions section says periodic serum level determinations of these drugs may be necessary.

Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)

Hyperlink: DailyMed

Phenytoin + Fluoxetine

Findings: Fluoxetine's label states there is potential for elevated phenytoin levels and clinical anticonvulsant toxicity when anticonvulsants such as phenytoin are used with fluoxetine. This is a DailyMed Drug Interactions label statement; it identifies phenytoin as an anticonvulsant for which elevated levels and toxicity are a potential concern during coadministration.

Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov

Hyperlink: DailyMed

Phenytoin + Haloperidol

Findings: Phenytoin coadministration accelerates haloperidol hepatic metabolism, increasing haloperidol clearance and reducing its plasma concentrations; this induction effect typically develops over 1–2 weeks and can persist for several weeks after phenytoin discontinuation. The source states this may reduce haloperidol effectiveness and risk breakthrough or worsening psychiatric symptoms, and recommends close monitoring when phenytoin is started or stopped, with possible haloperidol dose increase of 50–100% when phenytoin is added.

Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily)

Hyperlink: General web search (Tavily)

Phenytoin + Lamotrigine

Findings: Phenytoin decreases lamotrigine concentrations by approximately 40%. The DailyMed lamotrigine label identifies UGT enzymes as responsible for lamotrigine metabolism and lists phenytoin in the Drug Interactions section as decreasing lamotrigine concentration; no additional management recommendation for this specific pair is provided in the retrieved text.

Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily)

Hyperlink: DailyMed

Phenytoin + Methylphenidate

Findings: Methylphenidate may increase phenytoin serum levels according to the phenytoin DailyMed Drug Interactions table, where methylphenidate is listed among “other” interacting agents under drugs that may increase phenytoin concentrations. The excerpt does not provide an effect size or a specific management recommendation for methylphenidate beyond the label’s general goal of achieving optimal clinical outcome.

Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); General web search (Tavily)

Hyperlink: DailyMed

Phenytoin + Oxcarbazepine

Findings: Oxcarbazepine and phenytoin affect each other's monitored drug levels when coadministered: the label states that phenytoin may decrease blood levels of oxcarbazepine's active metabolite MHD, and that oxcarbazepine titration or dosage changes warrant monitoring phenytoin plasma levels. The DailyMed label recommends monitoring MHD during concurrent use with phenytoin and says Oxtellar XR dosage adjustment may be required; it also recommends monitoring phenytoin during Oxtellar XR titration or dosage modification because a phenytoin dose decrease may be required.

Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily)

Hyperlink: DailyMed

Phenytoin + Quetiapine

Findings: Phenytoin decreases quetiapine exposure by inducing CYP3A4; the label states that coadministration of quetiapine with phenytoin increased quetiapine mean oral clearance 5-fold. The DailyMed interaction section says quetiapine dose increases up to 5-fold may be required to maintain control of schizophrenia symptoms during chronic potent CYP3A4 inducer treatment, and the dose should be reduced back to the original level within 7–14 days after the inducer is discontinued.

Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)

POSITIVE IMPACT	NEGATIVE IMPACT
	Hyperlink: DailyMed **Phenytoin + Sertraline** **Findings:** Sertraline is listed in phenytoin's label among antidepressants that may increase phenytoin serum levels. The label states that adding or withdrawing such agents in patients receiving phenytoin may require phenytoin dose adjustment to achieve optimal clinical outcome. **Evidence:** DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** DailyMed **Phenytoin + Valproic Acid** **Findings:** Valproic acid is listed in the phenytoin label among antiepileptic drugs that may either increase or decrease phenytoin serum levels. The DailyMed Drug Interactions section states this as a label interaction for concomitant therapy; the retrieved text does not give a specific magnitude, mechanism, or pair-specific dose-adjustment recommendation. **Evidence:** DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** DailyMed

QUETIAPINE

POSITIVE IMPACT	NEGATIVE IMPACT
Quetiapine + Clobazam **Findings:** Discontinuation of clobazam and quetiapine was associated with clinical improvement in a 69-year-old man who developed MRI findings suggestive of central pontine myelinolysis despite normal serum sodium. This PubMed case report describes medication-induced CPM as a consideration for patients taking this combination who present with neurological symptoms, while preserving that the report presents a suspected association rather than a proven causal interaction. **Evidence:** External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** PubMed **Quetiapine + Methylphenidate** **Findings:** Adding quetiapine to ongoing OROS methylphenidate was associated with significant improvement in ADHD symptoms, aggression, and global functioning in adolescents who had not responded sufficiently to methylphenidate alone; at the end of combination treatment, 42% met all criteria for clinically significant improvement and 79% showed minimal aggression. In this prospective open-label study of participants aged 12–16 years, methylphenidate was titrated to 54 mg/day before 9 weeks of combined methylphenidate-quetiapine treatment; mild transient sedation occurred in about half of cases, and prior methylphenidate-associated weight loss was offset by weight gain during combination treatment. **Evidence:** External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** PubMed	**Quetiapine + Carbamazepine** **Findings:** Quetiapine's DailyMed label states: "prototype CYP3A4 inhibitors (e.g., ketoconazole, itraconazole, indinavir, ritonavir, nefazodone, etc.) and decreased by the prototype CYP3A4 inducers (e.g., phenytoin, carbamazepine, rifampin, avasimibe, St. John's wort etc.)." **Evidence:** DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed **Quetiapine + Phenytoin** **Findings:** Phenytoin decreases quetiapine exposure by inducing CYP3A4; the label states that coadministration of quetiapine with phenytoin increased quetiapine mean oral clearance 5-fold. The DailyMed interaction section says quetiapine dose increases up to 5-fold may be required to maintain control of schizophrenia symptoms during chronic potent CYP3A4 inducer treatment, and the dose should be reduced back to the original level within 7–14 days after the inducer is discontinued. **Evidence:** DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed **Quetiapine + Valproic Acid** **Findings:** General web evidence reports that quetiapine exposure could increase when valproic acid is given concomitantly, although the clinical significance is not known. The source cites a clinical study in 96 psychiatric patients showing a 77% increase in quetiapine plasma concentrations with concomitant valproic acid, while another producer-connected study found a 17% increase in quetiapine Cmax but no AUC change and a non-significant 11% reduction in valproic acid Cmax and AUC. It also notes case reports of cervical dystonia with quetiapine concentration above the therapeutic interval after valproic acid was added, and Parkinsonism during concomitant treatment. **Evidence:** General web search (Tavily) **Hyperlink:** General web search (Tavily)

SERTRALINE

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Sertraline + Atomoxetine** **Findings:** Sertraline hydrochloride may increase atomoxetine exposure because sertraline is a CYP2D6 inhibitor and atomoxetine is listed among CYP2D6 substrates in the label. The DailyMed interaction section recommends decreasing the dosage of a CYP2D6 substrate if needed during concomitant sertraline use, and notes that the substrate dose may need to be increased if sertraline is discontinued. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed **Sertraline + Bupropion** **Findings:** Bupropion inhibits CYP2D6 and can increase exposure to CYP2D6 substrates, including sertraline. The DailyMed label states that when sertraline is used

POSITIVE IMPACT	NEGATIVE IMPACT
	concomitantly with bupropion, it may be necessary to decrease the dose of the CYP2D6 substrate, particularly for drugs with a narrow therapeutic index. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed
	Sertraline + Carbamazepine **Findings:** General web evidence states that carbamazepine co-administered with sertraline accelerates hepatic breakdown of sertraline, reducing sertraline plasma concentrations; the induction effect is described as developing over 1–3 weeks and persisting for weeks after carbamazepine discontinuation. The source identifies the main clinical risk as reduced sertraline effectiveness, with possible worsening depression, anxiety, relapse, or treatment failure, and recommends close symptom monitoring. It also states that clinicians may need to increase the sertraline dose, typically by 25–50%, to maintain therapeutic effectiveness. **Evidence:** External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** General web search (Tavily)
	Sertraline + Citalopram **Findings:** Combined use of citalopram and sertraline has additive serotonergic activity and is associated with increased risk of serotonin syndrome, activation, sleep disturbance, and agitation. This pharmacodynamic interaction reference attributes the risk to overlapping serotonergic effects from both agents. **Evidence:** Pharmacodynamic interaction reference; External evidence (PubMed); General web search (Tavily)
	Sertraline + Escitalopram **Findings:** Combined escitalopram and sertraline have additive serotonergic activity, increasing the risk of serotonin syndrome as well as activation, sleep disturbance, and agitation. This pharmacodynamic interaction reference attributes the risk to both drugs acting as serotonergic agents when used together. **Evidence:** Pharmacodynamic interaction reference; External evidence (PubMed); ClinicalTrials.gov
	Sertraline + Fluoxetine **Findings:** Combined fluoxetine and sertraline has additive serotonergic activity and increases the risk of serotonin syndrome, activation, sleep disturbance, and agitation. The pharmacodynamic rule attributes this risk to combining serotonergic agents, so the clinically relevant implication is increased serotonergic adverse-effect risk when these drugs are used together. **Evidence:** Pharmacodynamic interaction reference; External evidence (PubMed); General web search (Tavily)
	Sertraline + Paroxetine **Findings:** Paroxetine and sertraline have additive serotonergic activity when combined, increasing the risk of serotonin syndrome as well as activation, sleep disturbance, and agitation. This pharmacodynamic interaction reference attributes the risk to both agents' serotonergic properties and identifies the clinically relevant consequence as increased serotonergic adverse-effect risk. **Evidence:** Pharmacodynamic interaction reference; General web search (Tavily)
	Sertraline + Phenytoin **Findings:** Sertraline is listed in phenytoin's label among antidepressants that may increase phenytoin serum levels. The label states that adding or withdrawing such agents in patients receiving phenytoin may require phenytoin dose adjustment to achieve optimal clinical outcome. **Evidence:** DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** DailyMed
	Sertraline + Venlafaxine **Findings:** Sertraline may increase venlafaxine exposure because the label states that sertraline is a CYP2D6 inhibitor and concomitant use with a CYP2D6 substrate, including venlafaxine, may increase exposure of the substrate. The label recommends decreasing the dosage of the CYP2D6 substrate if needed during concomitant sertraline use, and increasing the substrate dosage may be needed if sertraline is discontinued. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed

VALPROIC ACID

POSITIVE IMPACT	NEGATIVE IMPACT
Valproic Acid + Atomoxetine	**Valproic Acid + Carbamazepine**

POSITIVE IMPACT	NEGATIVE IMPACT
Findings: Methylphenidate or atomoxetine for 2 weeks also improved the prenatal VPA-induced decrease in dendritic spine density in the prefrontal cortex. The effects of these drugs on behavior **Evidence:** External evidence (PubMed) **Hyperlink:** PubMed	**Findings:** Carbamazepine can increase valproate clearance; the label states that hepatic enzyme-inducing drugs such as carbamazepine can increase valproate clearance and that carbamazepine can double the clearance of valproate. The DailyMed drug-interactions text recommends increased monitoring of valproate and concomitant drug concentrations, with dosage adjustment indicated when enzyme-inducing drugs are introduced or withdrawn. **Evidence:** DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** DailyMed
Valproic Acid + Duloxetine **Findings:** Duloxetine ameliorated valproic acid-induced hyperactivity, anxiety-like behavior, and social interaction deficits in a zebrafish model of ASD-like phenotypes. In this experimental study, embryonic valproic acid exposure produced persistent ASD-like behavioral and biochemical changes, and duloxetine was evaluated for rapid and persistent neuroprotective effects against those VPA-triggered deficits. **Evidence:** External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** PubMed	**Valproic Acid + Ethosuximide** **Findings:** Valproic acid has been reported to both increase and decrease ethosuximide levels when used concurrently. The DailyMed ethosuximide label recommends periodic serum level determinations for concurrently administered antiepileptic drugs because ethosuximide may interact with them. **Evidence:** DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** DailyMed
Valproic Acid + Methylphenidate **Findings:** Methylphenidate or atomoxetine for 2 weeks also improved the prenatal VPA-induced decrease in dendritic spine density in the prefrontal cortex. The effects of these drugs on behavior **Evidence:** External evidence (PubMed) **Hyperlink:** PubMed	**Valproic Acid + Phenytoin** **Findings:** Valproic acid is listed in the phenytoin label among antiepileptic drugs that may either increase or decrease phenytoin serum levels. The DailyMed Drug Interactions section states this as a label interaction for concomitant therapy; the retrieved text does not give a specific magnitude, mechanism, or pair-specific dose-adjustment recommendation. **Evidence:** DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** DailyMed
Valproic Acid + Nortriptyline **Findings:** Valproic acid's label states that dosage adjustment of amitriptyline/nortriptyline may be necessary when used concomitantly with valproic acid. The excerpt describes this as part of valproate's ability to affect the pharmacokinetics of other drugs by inhibiting metabolism or displacing protein binding, and recommends monitoring concomitant drug concentrations when enzyme-modifying drugs are introduced or withdrawn. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed	**Valproic Acid + Quetiapine** **Findings:** General web evidence reports that quetiapine exposure could increase when valproic acid is given concomitantly, although the clinical significance is not known. The source cites a clinical study in 96 psychiatric patients showing a 77% increase in quetiapine plasma concentrations with concomitant valproic acid, while another producer-connected study found a 17% increase in quetiapine Cmax but no AUC change and a non-significant 11% reduction in valproic acid Cmax and AUC. It also notes case reports of cervical dystonia with quetiapine concentration above the therapeutic interval after valproic acid was added, and Parkinsonism during concomitant treatment. **Evidence:** General web search (Tavily) **Hyperlink:** General web search (Tavily)

VENLAFAXINE

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Venlafaxine + Aripiprazole** **Findings:** The FDA-label-derived evidence states that no dosage adjustment is necessary for CYP2D6 substrates, including venlafaxine, when co-administered with aripiprazole. This is presented as an aripiprazole prescribing-label drug-interaction statement, with no additional strong caution language or specific adverse outcome stated for this pair. **Evidence:** DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** General web search (Tavily)
	Venlafaxine + Bupropion **Findings:** Bupropion inhibits CYP2D6 and can increase exposure to CYP2D6 substrates, including the antidepressant venlafaxine. The DailyMed label states that when venlafaxine is used concomitantly with bupropion, dose reduction of the CYP2D6 substrate may be necessary, particularly for drugs with a narrow therapeutic index. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed
	Venlafaxine + Haloperidol **Findings:** Venlafaxine's DailyMed label states: "(5.2) Cimetidine: Caution in patients with pre-existing hypertension, in elderly patients and patients with hepatic dysfunction. (7.2) Haloperidol: Increase in haloperidol AUC and C max . (7.4) Ketoconazole: Increase in venlafaxine and O-desmethylvenlafaxine AUC and C max ." **Evidence:** DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov **Hyperlink:** DailyMed

POSITIVE IMPACT	NEGATIVE IMPACT
	Venlafaxine + Lithium **Findings:** Venlafaxine's label states that a single 600 mg oral dose of lithium did not affect steady-state venlafaxine pharmacokinetics at 150 mg/day in 12 healthy male subjects, and O-desmethylvenlafaxine was also unaffected. The label also states that venlafaxine had no effect on lithium pharmacokinetics, but separately warns that serotonergic drugs including lithium have a potential for serotonin syndrome and advises careful patient observation. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) **Hyperlink:** DailyMed
	Venlafaxine + Sertraline **Findings:** Sertraline may increase venlafaxine exposure because the label states that sertraline is a CYP2D6 inhibitor and concomitant use with a CYP2D6 substrate, including venlafaxine, may increase exposure of the substrate. The label recommends decreasing the dosage of the CYP2D6 substrate if needed during concomitant sertraline use, and increasing the substrate dosage may be needed if sertraline is discontinued. **Evidence:** DailyMed label (Drug Interactions section); ClinicalTrials.gov **Hyperlink:** DailyMed

Known Limitations (Drug Interaction Flag Sheet):

- A pair with no finding from Flockhart, pharmacodynamic rules, PubMed, DailyMed, ClinicalTrials.gov, or Tavily is dropped from this report entirely, not shown as a negative/no-concern finding -- 'not listed here' means 'no evidence was found for this exact pair', never 'confirmed safe'.
- A DailyMed 'not mentioned in reviewed label section' result is never itself a finding -- it means this module checked the drug's real DRUG INTERACTIONS label section and the other drug's name was not there, not that no interaction exists; it only ever appears as supplementary context alongside whatever other evidence a pair has.
- Tavily (general web search) is queried directly and unconditionally alongside PubMed, DailyMed, and ClinicalTrials.gov for every pair here -- unlike elsewhere in this codebase (external_evidence_summary.py's question-answering paths), it is not gated behind the other sources returning no_results first. Every Tavily-sourced evidence entry is labeled 'General web search (Tavily)' so it is never mistaken for a dedicated medical/regulatory source, and it is always the lowest-priority candidate for a pair's primary Finding text.
- A source whose language does not clearly match an explicit positive-outcome or negative-outcome keyword never resolves a pair on its own -- this module falls through to the next candidate source in priority order instead, and if no candidate ever resolves clearly, the pair is dropped from the report entirely, the same as a pair with no evidence at all. There is no 'Needs Review' column here; most real titles/excerpts are topic descriptions, not verdicts, and this module never guesses which way an unclear one points.
- CPIC and ClinVar (gene-level sources) are not re-queried per pair here -- see the module docstring for why folding a single drug's own genotype-predicted finding into a pair's verdict here would conflate two different questions. Where this patient's genetics panel covers a screened drug, its real CPIC/ClinVar findings are in the Pharmacogenomics section, not duplicated here.
- Flockhart Cytochrome P450 Drug Interaction Table (curated-snapshot-2024) curated CYP-pathway data covers only a small hand-curated set of drugs; a pair outside that set relies on the other five sources, not a gap in this report.
- Positive/Negative classification is derived from keyword matching on publication titles, which can misread comparative-effectiveness study titles (where multiple drugs are studied together) as a verdict about one specific pair. This is a known limitation of title-only keyword classification, not a per-pair clinical conclusion.

Current Regimen — Curated-Source Coverage

MEDICINE	STATUS	MEANING	ACTION
Lamotrigine	RESOLVED	Curated CYP/pharmacokinetic reference data exists for this drug (Flockhart Cytochrome P450 Drug Interaction Table).	No additional action required beyond the current-current pair evidence below.
Levetiracetam (Keppra)	RESOLVED	Curated CYP/pharmacokinetic reference data exists for this drug (Flockhart Cytochrome P450 Drug Interaction Table).	No additional action required beyond the current-current pair evidence below.

Clinical inference: Absence of a resolved current-current interaction does not establish absence of interaction. Unresolved predicted-model coverage limits interpretation; the active medication list should be reassessed against a complete interaction source before any treatment change. The therapy-level cards below give the supporting, counter, and unresolved evidence behind that conclusion for each current medication, across pharmacokinetic, pharmacodynamic, pharmacogenomic, clinical, EEG, and laboratory sources -- not a pass/fail interaction check.

- SuperCYPsPred predicted pharmacokinetic evidence is not available locally for any drug in this regimen.
- No therapeutic drug concentration data is present in the source reports.
- No renal function (eGFR) data is present in the source reports.

Candidate / Proposed Medication-Pair Screening

No candidate or proposed medications are present in this patient's regimen; no candidate-pair screening applies.

Lamotrigine EFFECTIVENESS INCOMPLETE

Therapeutic evidence present, but clinical effectiveness remains incomplete.

Supporting Evidence

- CBC status across the observation window: stable within reference ranges; no documented hematological toxicity. — digital_twin_state
- No documented drug toxicity for this agent in the clinical-notes source. — clinical_inference.medication_response
- SCN1A (CT): efficacy — findings_by_therapeutic_class

Counter-Evidence

- Breakthrough seizures persisted on the current regimen, with a recorded peak of 11 episodes on 2026-02-15. — clinical_inference.seizure_analysis
- ABCB1 (GG): reduced efficacy — findings_by_therapeutic_class
- CYP3A4 (TT): reduced efficacy — findings_by_therapeutic_class

Clinical Impression

4 supporting, 5 counter, and 13 unresolved evidence item(s) were identified for Lamotrigine. Clinical-notes inference: Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.

Recommended Monitoring

Continue EEG-derived seizure-risk monitoring. | Continue periodic CBC and safety-laboratory monitoring. | Continue seizure frequency, duration, and clustering tracking. | Review the priority safety flags in the Pharmacogenomics section before adding or changing an interacting medication.

Levetiracetam (Keppra) EFFECTIVENESS INCOMPLETE

Therapeutic evidence present, but clinical effectiveness remains incomplete.

Supporting Evidence

- CBC status across the observation window: stable within reference ranges; no documented hematological toxicity. — digital_twin_state
- No documented drug toxicity for this agent in the clinical-notes source. — clinical_inference.medication_response
- SCN1A (CT): efficacy — findings_by_therapeutic_class

Counter-Evidence

- Breakthrough seizures persisted on the current regimen, with a recorded peak of 11 episodes on 2026-02-15. — clinical_inference.seizure_analysis
- Seizure clustering was documented during the observation window. — clinical_inference.seizure_analysis
- EEG synthesis remained consistent with active epilepsy (EEG-derived risk: Elevated; future seizure probability: Elevated). — digital_twin_state

Clinical Impression

3 supporting, 3 counter, and 7 unresolved evidence item(s) were identified for Levetiracetam (Keppra). Clinical-notes inference: Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.

Recommended Monitoring

Continue EEG-derived seizure-risk monitoring. | Continue periodic CBC and safety-laboratory monitoring. | Continue seizure frequency, duration, and clustering tracking. | Review the priority safety flags in the Pharmacogenomics section before adding or changing an interacting medication.

Limitations of this screen: This assessment is clinical decision support, not an automated prescribing system; no dose, start, or stop decision is made or implied here. No proposed/candidate medication list is present in the source data, so only the current regimen's pairwise and therapy-level evidence is evaluated. Curated pharmacokinetic (CYP) coverage is limited to a small bundled reference table; a drug not listed there is reported as unresolved, not as free of interactions. Therapy assessments also draw on the live external-evidence layer (see Known Limitations under Pharmacogenomics, above, for its source-by-source scope and caveats).

07 Pharmacogenomics

76 variants across 47 medications · reported 10 Sep 2026

Pharmacogenomic Panel Summary

CYP2D6 **GG**	CYP2C19 **GA**	CYP2C9 ***1/*3 - Intermediate Metabolizer**	CYP2B6 ***1/*6 - Intermediate Metabolizer**	CYP3A5 ***3/*3 - Poor Metabolizer**
CYP1A2 **AA**	CYP3A4 **TT**	UGT1A4 ***1/*3A - Decreased Function**	POR ***1/*28 - Decreased Function**	

Interpretive method: a pharmacogenomic association is weighed against this patient's own observed clinical response, laboratory, and EEG evidence -- it is never used as an isolated prescribing instruction, and a favorable genotype cannot override documented toxicity or persistent symptoms. The table below shows the single most clinically significant finding per current-regimen drug; the full therapy-level assessment -- this finding combined with clinical, EEG, laboratory, and external-database evidence -- is presented in Section 06, Drug Interactions.

Patient-Specific PGx Findings

THERAPY	GENE/GENOTYPE	PATIENT-SPECIFIC INTERPRETATION	CLINICAL RELEVANCE
Lamotrigine	CYP3A4 (TT)	Genotype is associated with a reduced-efficacy response to this therapy.	Moderate relevance; flagged as clinically significant in the source panel (citation: PMID:28343093).
Levetiracetam	SCN1A (CT)	Genotype is associated with a favorable efficacy response to this therapy.	Supportive relevance; flagged as clinically significant in the source panel (citation: PMID:28753467).

External Evidence Verification

Database evidence only, not patient-verified: each block below is a live public-database lookup for the same gene/drug pair as the patient-specific finding in Table 1 above -- it establishes general scientific plausibility for that pair, not a fact confirmed for this individual patient. An unresolved result means this exact query returned nothing, or failed, at the time checked -- it is not evidence that no relevant data exists, and must not be read as a clean or reassuring negative.

- **CYP3A4 → Lamotrigine** PARTIALLY VERIFIED
  - CPIC: Unresolved. ClinVar: Uncertain significance.
  - PubMed: PMID 21635243 found. Overall: Partially resolved (2 of 3 sources).
- **SCN1A → Levetiracetam** PARTIALLY VERIFIED
  - CPIC: Unresolved. ClinVar: Uncertain significance.
  - PubMed: PMID 39200163 found. Overall: Partially resolved (2 of 3 sources).

Provenance

- **CYP3A4 → Lamotrigine:** ClinVar — Accession VCV004885026 (70 total gene-wide records) · LIVE · Retrieved 11 Sep 2026; PubMed — PMID 21635243 · LIVE · Retrieved 11 Sep 2026. Note: Table 1's citation is from the patient's own PGx report; this is an independent live PubMed search, may differ.
- **SCN1A → Levetiracetam:** ClinVar — Accession VCV004887952 (5406 total gene-wide records) · LIVE · Retrieved 11 Sep 2026; PubMed — PMID 39200163 · LIVE · Retrieved 11 Sep 2026. Note: Table 1's citation is from the patient's own PGx report; this is an independent live PubMed search, may differ.

Evidence Scope: live-verified via ClinVar, CPIC, DailyMed, PubMed, and ClinicalTrials.gov only. The rest of the NeuroTwin resource catalog (PharmGKB, DrugBank, Reactome, STRING, GTEx, and others) is out of scope for this live-lookup layer and is not reflected anywhere in this table — see Known Limitations below.

Known Limitations (External Evidence):

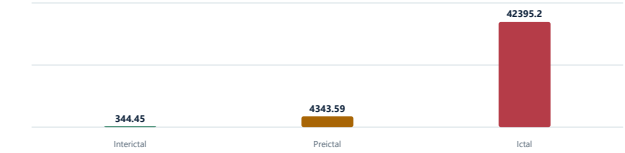
- This module queries exactly five public APIs (PubMed, ClinVar, DailyMed, CPIC, ClinicalTrials.gov); it does not cover the rest of the NeuroTwin resource catalog (Reactome, STRING, GTEx, RCSB PDB, etc.) -- those remain out of scope for this module and must not be inferred from it.
- A 'no_results' status means the API returned zero matching records for the exact query issued; it is not evidence of absence, and must not be reported as a negative finding.
- Responses may be served from the on-disk cache in reports/external_evidence/.cache/ rather than a fresh call; each record's 'from_cache' and 'retrieved_at' fields state which. Pass --refresh to force live calls.

08 EEG

Longitudinal interictal / preictal / ictal signal characteristics

35	12	11	12
Total Recordings	Interictal	Preictal	Ictal

Signal Variance Midpoint by Phase (μV²)



EEG-derived risk: Elevated

Future seizure probability: Elevated

Dominant feature: Signal Variance

Phase Analysis

PHASE	STATE	ACTIVITY	VARIABILITY
Interictal	Baseline	Stable	Low
Preictal	Transition	Increasing Instability	Moderate-High
Ictal	Seizure	Hypersynchronous	Very High

PHASE	AMPLITUDE RANGE	STD DEV RANGE	VARIANCE RANGE
Interictal	-89; 99	11.68; 23.5	136.44; 552.46
Preictal	-345; 440	14.48; 92.07	209.71; 8477.48
Ictal	-504; 784	52.5; 286.42	2756.47; 82033.96

Longitudinal EEG recordings demonstrate a consistent transition from stable interictal activity to preictal electrical instability followed by high-amplitude ictal seizure activity. Findings are consistent with the dataset's interictal/preictal/ictal phase taxonomy, indicating reproducible seizure-related signal evolution; this is a signal-classification observation, not a clinician-verified diagnosis.

09 MRI / CT

Structural imaging: neuroimaging key findings

CT — 6 studies, 12 Jan – 29 Jun 2026

DATE	IMPRESSION
12 Jan 2026	No acute intracranial abnormality is evident on this single-image model-assisted review.
16 Feb 2026	No gross acute abnormality is evident on this limited skull-base level image; assessment is limited by slice level and image quality.
11 Mar 2026	No acute intracranial abnormality is evident on this single-image model-assisted review.
20 Apr 2026	No acute intracranial abnormality is evident on this high-convexity image; ventricular assessment is not available at this slice level.
14 May 2026	No acute intracranial abnormality is evident on this limited-quality near-vertex image.
29 Jun 2026	No acute intracranial abnormality is evident. Incidental bilateral choroid plexus calcifications are present.

Across the available CT image files, no gross acute intracranial hemorrhage, midline shift, or mass effect is evident on this model-assisted image-only review. Direct progression assessment is limited because the files represent single slices with different slice levels, resolutions, and image quality. This output is not a radiologist-verified diagnostic report.

MRI — Metadata-supported structural stability, diagnostic interpretation not available

FINDING	STATUS	DETAIL
Brain Morphology	NOT DIRECTLY ASSESSED	The provided files include volumetric MRI metadata, but no radiologist interpretation or morphometric segmentation was available for direct morphology assessment.
Brain Volume	STABLE WITHIN AVAILABLE METADATA	T1w nonzero volume range: 8387642 to 8777268 mm3 (mean 8641065 mm3; range 4.5%). FLAIR nonzero volume range: 6022385 to 6625690 mm3 (mean 6231344 mm3; range 9.7%). These are preprocessing-derived nonzero voxel volumes, not formal brain volumetry.
Tissue Signal Characteristics	QUANTITATIVELY SUMMARIZED, NOT DIAGNOSTICALLY INTERPRETED	FLAIR mean intensity 28.76 to 32.27; standard deviation 47.57 to 54.16; T1w mean intensity 99.82 to 105.56; standard deviation 153.13 to 162.05. Intensity statistics are useful for preprocessing consistency but are not diagnostic tissue-signal interpretation.

Not directly assessed from the available imaging metadata (this means "not evaluated," not "abnormal"): White Matter, Hippocampus, Cortical Structure, Mass Lesion, Hemorrhage, Infarction.

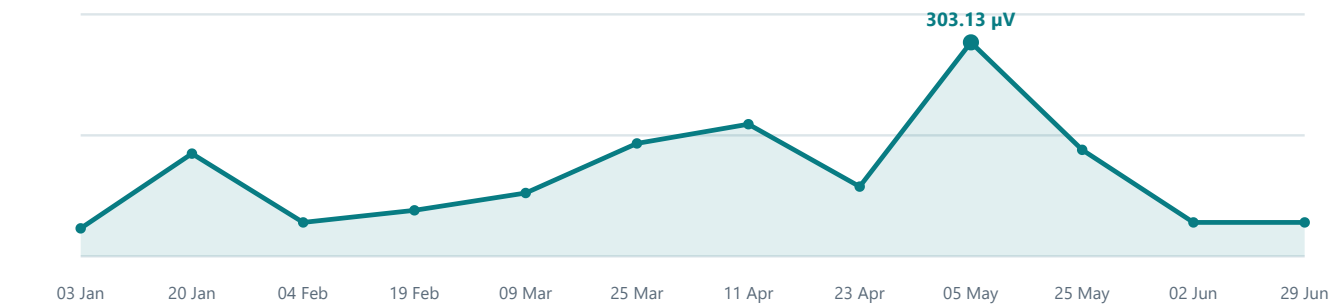
Stable by available preprocessing metadata. 6 MRI series were available across 3 study identifiers, including 3D T1-weighted MRI, T2-FLAIR MRI. Preprocessed metadata show consistent 3D acquisition characteristics and no metadata-level evidence of major global structural progression. A radiology report, lesion segmentation, hippocampal volumetry, and cortical thickness analysis were not available, so subtle pathology cannot be excluded from these metadata alone.

10 ECG

single channel ("ECG SD") in every recording

12	99h 11m	256
Recordings	Total Duration	Sampling Rate (Hz)

Amplitude Std Dev by Recording Date (µV)



- All 12 recordings use a single channel sampled at 256 Hz, so statistics are directly comparable across sessions.
- Recording duration varies considerably, from about 1h 11m (2026-04-11) to about 22h 27m (2026-03-25).
- The 2026-06-02 and 2026-06-29 recordings have matching percentile values and near-identical histogram shapes, suggesting very similar signal characteristics or a data duplication between the two sessions.
- Standard deviation of amplitude ranges from about 82.6 uV (2026-01-03) to about 303.1 uV (2026-05-05), indicating substantially different signal variability/noise levels between sessions.
- Peak-to-peak amplitude (max minus min) ranges from roughly 2,483 uV (2026-05-25) to about 3,577 uV (2026-03-25).

This report is derived strictly from the statistical fields present in the source dataset. It does not represent a clinical or diagnostic ECG interpretation. Any assessment of cardiac rhythm, heart rate, or abnormality requires review of the underlying waveform by a qualified clinician or an appropriately validated diagnostic algorithm, neither of which is reflected in this report.

11 CBC / Labs

12 reports, 01 Jan – 30 Jun 2026

Overall hematological status: Stable within reference ranges. No CBC parameter abnormality was identified against the report-provided reference ranges across the observation period.

PARAMETER	FIRST	LATEST	RANGE	TREND
Hemoglobin (Hb)	14.4	15.1	13.5 – 15.9	increasing
Total RBC count	5.0	5.1	4.8 – 5.3	increasing
Packed Cell Volume (PCV)	44.7	43.5	42.2 – 48.4	decreasing
Mean Corpuscular Volume (MCV)	90.0	97.8	88.4 – 98.8	increasing
MCH	27.6	29.9	27.6 – 31.1	increasing
MCHC	33.1	33.0	32.9 – 34.3	decreasing
RDW	12.1	13.5	12.1 – 13.5	increasing
Total WBC count	5761.0	9630.0	5476.0 – 9973.0	increasing
Neutrophils	58.0	54.0	52.0 – 61.0	decreasing
Lymphocytes	22.0	29.0	22.0 – 34.0	increasing
Eosinophils	4.0	4.0	1.0 – 5.0	stable
Monocytes	3.0	3.0	1.0 – 8.0	stable
Basophils	1.0	1.0	0.0 – 2.0	stable
Platelet Count	251725.0	209427.0	189551.0 – 373102.0	decreasing

Patient-reported vs. measured discrepancy: the 30 Jun 2026 questionnaire records patient recollections of White Blood Cell (WBC) count baseline trend from your latest Complete Blood Count (CBC): Mild Leukopenia (Borderline low); Platelet Count profile stability from recent blood evaluations: Borderline Thrombocytopenia (Low); Liver Function Test (LFT) - SGOT/AST status: Mild Elevation (40-100 U/L); measured CBC values across all 12 reports in this period remained within their reference ranges throughout (see table below). The discrepancy likely reflects patient recall/perception rather than a measured lab abnormality.

12 Sleep, Cognition & Quality of Life

Sleep

TREND	Sleep remained variable across visits
AVERAGE	6.2 hours per night across printed assessments

Daytime naps needed
Difficulty falling back asleep
Fatigue level through the day
Waking up unrefreshed

Cognition

MEMORY	Stable on clinician review, with low to intermittent patient-reported memory difficulty
ATTENTION	Variable patient-reported concentration and attention symptoms
LANGUAGE	Mild word-finding difficulty was noted but not described as worsening
MENTAL CLARITY	Improved mental clarity was reported at the latest visit

Quality of Life

DAILY FUNCTION	Independence in daily activities was repeatedly normal, intact, or stable
SOCIAL FUNCTION	Intermittent social withdrawal or isolation symptoms were reported
CLINICAL INTERPRETATION	Quality of life remains affected by seizures, sleep, triggers, and medication tolerability, despite generally stable outpatient status.

Latest Questionnaire Domain Scores (0-100)



13 Digital Twin Risk Dashboard

Overall clinical risk: Moderate

Future Seizure Risk: Moderate	Status Epilepticus Risk: Low To Moderate	Hospitalization Risk: Low To Moderate	Cognitive Decline Risk: Low
Medication Toxicity Risk: Low To Moderate	Functional Decline Risk: Low	EEG-Derived Seizure Risk: Elevated	Genetic Safety Flags: 10 Flags, 7 High

State Vector (Health Metric Scores, 0-100)



Each domain score is the mean of clinician-documented per-visit ratings across the clinical-note window; overall health score is the mean of the other domains. Derived digital-twin summary metrics, not a direct lab/imaging measurement.

14 Clinical Action Plan

Priority Actions (Pharmacogenomics)

HIGH **Avoid or use with heightened monitoring**

See priority_safety_flags for the specific gene-drug pairs (ABCB1 (CC), BDNF (TC), CREB1 (AA), FKBP5 (TC), GRIA3 (A), HTR1B (TA), METTL21A (AA)).

MODERATE **Dose with caution / consider therapeutic drug monitoring**

See metabolizer_profile for reduced-function genes (CYP2C9, CYP2B6, CYP3A5, UGT1A4, POR).

INFORMATIONAL **Contextualize**

Interpret alongside actual clinical history, current medication list, and labs -- this panel alone does not establish a diagnosis.

Recommended Actions

- continue the documented antiseizure medication regimen unless changed by the treating clinician
- maintain the documented plan for consistent sleep schedule and trigger avoidance
- continue seizure and aura tracking across follow-up windows
- continue periodic CBC, liver function, and antiseizure drug-level safety monitoring when scheduled
- reassess at the planned follow-up interval

Monitoring Priorities

- seizure frequency by type
- seizure clustering
- average and longest seizure duration
- post-ictal recovery time
- aura pattern and severity
- sleep duration and sleep fragmentation
- missed medication doses and trigger exposure
- dizziness and other medication side effects
- CBC, liver function, and antiseizure drug-level tracking
- word-finding, memory, and attention symptoms
- Signal Variance
- Amplitude Dynamics
- Electrical Stability
- Preictal Detection
- Seizure Frequency
- Temporal Disease Progression

MRI Follow-Up

- Perform formal radiology review of the MRI image volumes.
- Run automated brain segmentation and longitudinal registration across the three study identifiers.
- Perform hippocampal volumetry because epilepsy-related mesial temporal abnormalities cannot be excluded from metadata alone.
- Run FLAIR white matter lesion analysis if clinically relevant.
- Compare future MRI studies using the same T1w and FLAIR acquisition strategy when possible.
- Correlate MRI findings with seizure semiology and neurological examination.
- Review MRI together with EEG findings for seizure-focus localization.
- Repeat or escalate imaging review if new neurological deficits, worsening seizures, or new focal symptoms occur.

Medication Reconciliation

- Current regimen reconciled and checked against curated reference pharmacology and this patient's own pharmacogenomic findings (Section 07). Predicted-model (SuperCYPsPred) coverage is still unresolved, so a live, comprehensive DDI screen is recommended before adding or changing any interacting medication.